

Hybrid Quantum-Classical Hierarchical Actor-Critic for Sum-Rate Maximization in IRS-Assisted Grouped RSMA Systems

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Abstract

Integrated satellite-terrestrial networks are impaired by building blockage of the direct path and by the worst-user bottleneck that a shared common stream imposes. We consider an IRS-assisted grouped rate-splitting multiple access downlink and jointly optimize four coupled decision variables, namely IRS-user link assignment, phase-shift configuration, transmit power allocation and common-rate distribution, under discrete phase quantization, blockage and imperfect channel state information. Their strict sequential dependency motivates a hybrid quantum-classical hierarchical actor-critic framework in which a variational quantum circuit resolves the combinatorial assignment while three classical actors govern the remainder. A direct state encoding would grow the register with network size, so an autoencoder holds it at eight qubits across both scales. Against a per-state alternating-optimization solver the framework gains roughly 30% on the QoS-penalized objective at five users and one IRS, narrowing to seed-variance parity at ten users and two IRSs. The quantum actor matches an equivalent classical network within seed variance at 6 to $9\times$ fewer inference parameters, and every learned policy issues three rate-kernel invocations per decision regardless of user count, against 482 and 896 for the solver. The advantage accordingly resides in parameter efficiency and amortized inference rather than in achievable rate.

Index Terms

Hierarchical actor-critic (HAC), integrated satellite-terrestrial networks (ISTN), intelligent reflecting surface (IRS), quality of service (QoS), rate-splitting multiple access (RSMA), sum-rate maximization, variational quantum circuit (VQC).

I. INTRODUCTION

A. Background

Integrated satellite-terrestrial networks (ISTNs) extend coverage far beyond the reach of fixed ground infrastructure, and falling launch costs have made constellation-scale deployment practical [1]–[3]. The orbital link is nonetheless physically fragile. The extended path incurs substantial free-space path loss, the atmosphere adds rain attenuation and tropospheric scintillation [4], and in densely built environments buildings shadow the direct satellite-to-user link, depriving the affected terminal of a reliable primary propagation path.

What the geometry leaves behind is addressed by two techniques, the first of which confronts the worst-user bottleneck, whereby the terminals in the most adverse conditions constrain the aggregate rate of the entire served population [5]. Rate-splitting multiple access (RSMA) relaxes that bottleneck by decomposing each message into a common component, decoded by every receiver in the designated set and then removed by successive interference cancellation [6], [7], and a private component decoded independently, a grouped extension bounding the decodable common rate by the weakest member of a subgroup rather than by the globally worst user [8]. Rate splitting cannot by itself remedy an obstructed path, for which an intelligent reflecting surface (IRS) [9] supplies an alternative route. Its passive elements each impose an independently controllable phase shift, quantized in practice to a few discrete levels, and configuring those shifts renders the reflected contributions coherent at an intended receiver, establishing a virtual line-of-sight link without additional radiated power. Under grouped RSMA each active surface is the reflective backbone of one user group, as illustrated in Fig. 1.

Operating such a network requires four coupled decisions at every scheduling interval, namely the assignment of users to surfaces, the phase configuration of each surface, the division of the transmit budget and the apportionment of the group common rate. Together they form a mixed-integer non-convex problem admitting no tractable closed form, for which the established remedy is to decompose the problem and alternate between the resulting subproblems [10], [11]. That remedy is priced per channel realization, the solver retaining nothing between invocations, so that every realization pays the iterative cost anew and a link whose geometry follows user mobility pays it at the rate at which that geometry changes [12].

Reinforcement learning inverts the arrangement. A policy trained once maps an observed state directly

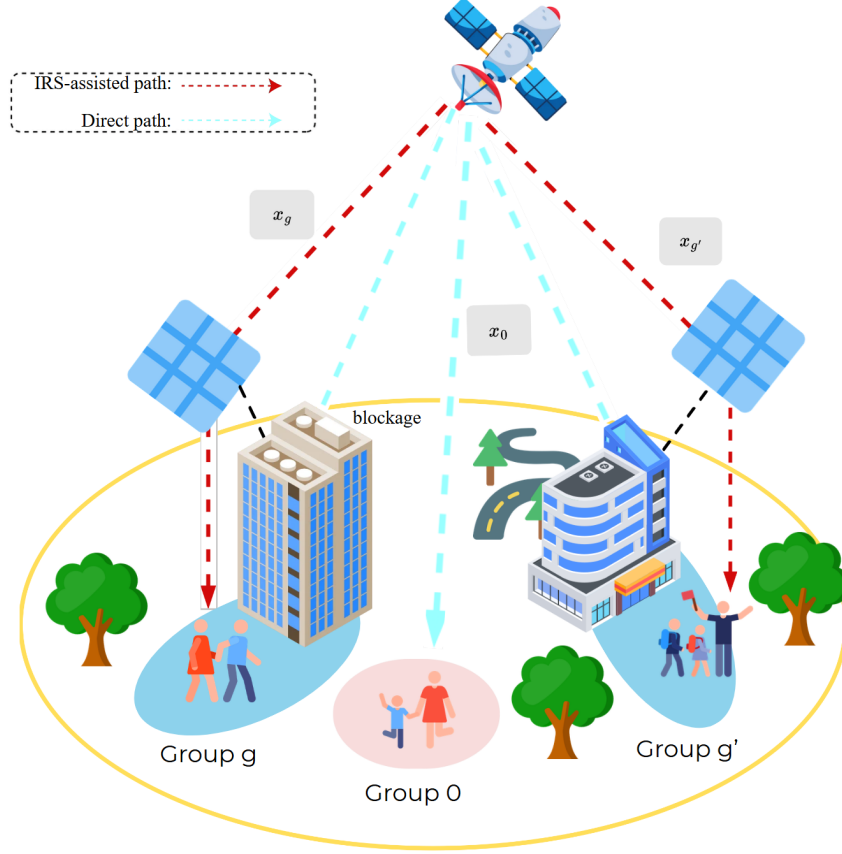


Fig. 1: Proposed IRS-assisted grouped RSMA system in an ISTN downlink. The satellite simultaneously serves $G + 1$ groups: Group 0 users with unobstructed satellite links receive the direct signal x_0 , while Group g and Group g' users blocked by buildings are served via dedicated IRS panels carrying reflected signals x_g and $x_{g'}$. Each IRS serves its assigned group through the phase-shift configuration Φ_m , chosen so that its reflected contributions combine coherently at those users.

onto an allocation, discharging the optimization burden offline and answering each realization with a single forward pass [13], [14]. The four variables are not independent but form a strict sequential dependency, the assignment fixing the grouping that governs the phase design, the phase design shaping the effective channels that enter the power allocation, and the resulting budget delimiting the feasible common-rate split. That ordering motivates a hierarchical actor-critic decomposition [15], [16], a high-level actor committing to the discrete assignment while dedicated low-level actors resolve the remainder in sequence. The discrete complexity concentrates in the assignment, the number of feasible user-to-surface associations growing exponentially in the user population. A variational quantum circuit (VQC) represents a distribution over that space with a compact set of trainable parameters [17], and because its input would otherwise scale with both the user population and the panel count and so exceed a near-term register [18], an autoencoder [19] learns the latent embedding on which the assignment actually depends, following [20].

TABLE I: Comparison of existing works on IRS/RIS-assisted communication systems

Ref.	Access	Blockage	CSI	Prop. Path	IRS/RIS			RL	QC	QoS	Obj.
					Scale	Type	Phase				
2019 [21]	MU-BF	–	Perf.	Multi	Single	Passive	Cont.	–	–	Constr.	Min-power
2022 [22]	SDMA	✓	Imperf.	Multi	Single	Passive	Cont.	–	–	–	Sum-rate
2023 [23]	TDMA	✓	Perf.	Multi	Single	Passive	Cont.	–	–	–	WSR
2023 [12]	RSMA	–	Perf.	Multi	Single	STAR	Cont.	✓	–	Constr.	Sum-rate
2025 [24]	H-NOMA	–	Perf.	Single	Single	Passive	Cont.	–	–	Constr.	TA-SR
2025 [25]	RSMA	–	Imperf.	Multi	Single	Passive	Disc.	–	–	Constr.	WSR
2025 [26]	RSMA	–	Perf.	Single	Single	Passive	Disc.	–	–	Constr.	WSR
2025 [27]	MU-BF	–	Perf.	Multi	Single	Passive	Cont.	–	–	–	WSR
2025 [28]	NOMA	–	Perf.	Multi	Single	Passive	Cont.	✓	–	Penalty	ASR
2025 [29]	RSMA	✓	Perf.	Single	Multi	Passive	Cont.	–	–	Constr.	Sum-rate
2026 [30]	RSMA	–	Imperf.	Multi	Multi	Passive	Cont.	–	–	Constr.	WSR
2026 [31]	RSMA	–	Imperf.	Multi	Single	Active	Cont.	–	–	Constr.	Sum-rate
2026 [8]	RSMA	–	Perf.	Single	Single	Passive	Cont.	–	–	Constr.	Sum-rate
Proposed	G-RSMA	✓	Imperf.	Multi	Multi	Passive	Disc.	✓	✓	Penalty	Sum-rate

Table I places the setting against the prior art. The entries differ in access scheme, panel count, phase resolution and CSI assumption, and the configuration studied here sits at the demanding end of each. That a per-realization solver adapts poorly to a changing environment is itself an established argument [12]. What the table does not record, for the model-based and the learned entries alike, is what producing one allocation costs. That is the question this paper takes up, and it is answered in units that do not depend on the machine the policy runs on.

B. Contributions

The framework developed here addresses that gap from the standpoint of decision cost rather than of achievable rate. It resolves the four variables with a single hierarchical policy that is trained once and thereafter commits to an allocation in one forward pass, and it places a variational quantum circuit at the head of the hierarchy, where the discrete complexity is concentrated. The contributions are summarized as follows.

- We formulate sum-rate maximization for a multi-IRS-assisted grouped RSMA downlink in ISTN, jointly optimizing IRS-user link assignment, phase-shift configuration, transmit power allocation and common-rate distribution as a sequential dependency chain, and balancing aggregate rate against per-user QoS under discrete phase quantization, building blockage and imperfect CSI.
- We propose the HQC-HAC framework, which exploits that dependency. A quantum actor built on a variational quantum circuit resolves the combinatorial IRS-user assignment at the high level,

three classical actors resolve the phase, power and common-rate decisions in sequence, and entropy regularization with an annealing schedule sustains exploration across all levels.

- We adapt the AE-VQC pairing of [20] by structuring what the circuit receives. An affinity feature map removes the dependence on the reflecting-element count N , and a dual-branch autoencoder compresses the result into a latent whose two blocks load onto the user and IRS registers indexed by the readout observables, so the compression preserves the user–IRS structure.
- We evaluate two instantiations differing only in the assignment actor and report what a decision costs in units independent of the implementation. The quantum instantiation matches its classical counterpart to within the spread across training seeds while carrying substantially fewer inference parameters, and the invocation count of every learned policy is constant in the user population where that of the solver grows. Ablations attribute this to the compression rather than to the register size, the observable set or the entangling structure, placing the contribution in parameter efficiency and amortized inference rather than in achievable rate.

II. RELATED WORKS

A. IRS-Assisted Grouped RSMA for Sum-Rate Optimization in ISTNs

The foundational framework for IRS-assisted wireless networks was established by Wu and Zhang [21], who demonstrated that jointly optimizing active transmit beamforming at the access point and passive phase shifts at the surface substantially reduces the transmit power required to serve a multiuser MISO downlink. Aligning those shifts so that the reflected signal combines coherently with the direct path at the intended receiver establishes a virtual line-of-sight link, circumventing physical obstruction without any additional radiated energy.

The broader body of work summarized in Table I leaves the same axes open. Ten of the twelve entries deploy a single reflecting panel, so the reflected coverage cannot follow spatially dispersed user groups, the exceptions being [29], [30]. Eight presume perfect CSI, robust designs being confined to [22], [25], [30], [31], and continuous phase control is near-universal, with quantization appearing only in [25], [26]. Learning-based optimization is rare, adopted in [12], [28], and no entry employs a quantum policy representation. Blockage enters the propagation model in only three cases [22], [23], [29].

Building upon this beamforming paradigm, Tan et al. [8] extended IRS-assisted transmission to satellite downlink communications by proposing a flexible grouped RSMA scheme for sum-rate maximization in ISTNs. In their system, the K ground users are dynamically partitioned into two disjoint groups based on real-time channel conditions and potential line-of-sight obstructions: group $\mathcal{K}_1$ users are exclusively served via the satellite-IRS-user reflected link, while group $\mathcal{K}_2$ users communicate through the direct satellite-user link. The decodable common rate within each group is thereby bounded by the weakest member of that subgroup rather than by the globally worst-performing user, which substantially alleviates the worst-user bottleneck inherent in flat RSMA designs. Their joint problem spans satellite precoding beamforming, IRS phase-shift configuration, rate-splitting allocation and binary group selection, and was decomposed into four subproblems solved respectively by SDR with polyblock outer approximation, Riemannian manifold conjugate gradient descent on the phase manifold, alternating optimization for rate allocation, and simulated annealing for the binary assignment, the joint solution being recovered iteratively.

Despite reporting approximately 29% sum-rate gain over the IRS-assisted RSMA baseline, the framework of [8] carries several limitations pertinent to realistic deployment. The optimization presupposes perfect and instantaneous CSI at the satellite, whereas practical satellite links are susceptible to estimation errors arising from channel aging, Doppler effects, feedback delay, and limited feedback bandwidth. Only a single IRS panel is considered, which is insufficient to extend reflected coverage across spatially dispersed user groups. Furthermore, the iterative model-based solver must be re-run from scratch for every channel realization, and the framework provides no mechanism for adaptive policy adjustment under dynamic channel conditions or user mobility. The proposed HQC-HAC framework addresses these deficiencies by optimizing IRS-user link assignment across multiple IRS panels under imperfect CSI, building blockage, and dynamic user mobility, replacing model-based sub-problem decomposition with a hierarchical reinforcement learning approach whose variational quantum policy representation manages the combinatorial complexity of multi-IRS user assignment.

B. Entropy-Regularized Hierarchical Reinforcement Learning and Hybrid Quantum-Classical Policies

Hierarchical actor-critic (HAC) methods decompose complex decision problems into multiple levels of abstraction, with high-level actors committing to sub-goals that low-level actors subsequently fulfill [15], [32]. This temporal and functional decomposition has been demonstrated to accelerate learning in environ-

ments with large composite action spaces, including IRS-assisted wireless resource management, where jointly optimizing discrete and continuous variables over extended interaction horizons is otherwise costly to learn directly [16]. Entropy regularization sustains policy diversity during training, the maximum-entropy objective of soft actor-critic [33] penalizing premature collapse to a deterministic policy.

Hybrid quantum-classical reinforcement learning extends this paradigm by replacing or augmenting classical policy networks with variational quantum circuits (VQCs) [17], [34]. VQCs model nonlinear policy mappings through parameterized quantum state transformations with comparatively compact trainable representations, which motivates their study for high-dimensional combinatorial decisions and has prompted direct parameter-based comparisons against neural networks [35]. However, directly encoding high-dimensional wireless system states into a quantum register is impractical given current hardware constraints [18]. Nagy et al. [20] addressed this by jointly training an autoencoder (AE) with a VQC-based policy, where the AE compresses the original observation into a low-dimensional latent space suitable for quantum encoding. Motivated by these complementary contributions, this work embeds an AE-VQC quantum actor within an entropy-regularized HAC structure, applying it to the combinatorial IRS-user assignment problem in IRS-assisted grouped RSMA over ISTN downlinks.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. Network and Association Model

We consider an IRS-assisted grouped rate-splitting multiple access (RSMA) downlink communication system in an integrated satellite-terrestrial network (ISTN). In the considered system, a satellite serves a set of ground users with the assistance of intelligent reflecting surfaces (IRSs), which are deployed on high buildings to enhance the wireless propagation environment and support users experiencing weak or blocked direct satellite-to-user links. The satellite is assumed to act as the central controller, which determines the transmission strategy, resource allocation, and IRS-assisted communication mode according to the observed channel conditions and user requirements.

Let $\mathcal{K} = \{1, 2, \dots, K\}$ denote the set of downlink ground users, where K is the total number of users. The satellite transmits one independent message to each user, the message intended for user $k \in \mathcal{K}$ being denoted by s_k , so that the scheduled message set is $s = \{s_1, \dots, s_K\}$.

We consider M IRSs deployed on the rooftops of buildings to assist the satellite downlink transmission. Let $\mathcal{M} = \{1, 2, \dots, M\}$ denote the set of IRSs. Each IRS is equipped with N passive reflecting elements, which can independently adjust the phase shift of the incident signal. The IRSs are assumed to be connected to the central controller, and their reflection coefficients are configured according to the channel conditions and user association decisions. For IRS $m \in \mathcal{M}$, the reflection matrix is defined as

$$\Phi_m = \text{diag} \left(e^{j\theta_{m,1}}, e^{j\theta_{m,2}}, \dots, e^{j\theta_{m,N}} \right), \quad (1)$$

where $\theta_{m,n}$ denotes the phase shift of the n -th reflecting element at IRS m . Practical surfaces resolve the phase only to a finite number of levels, so with B quantization bits each element selects its shift from the uniformly spaced set [25], [26]

$$\mathcal{Q}_\theta = \left\{ \theta^{(q)} = \frac{2\pi q}{2^B} \mid q = 0, 1, \dots, 2^B - 1 \right\}. \quad (2)$$

To determine the transmission mode of each user, we define an IRS-selection variable for every user $k \in \mathcal{K}$. Since the considered system includes M IRSs, each user can either be served through the direct satellite-to-user link or through one of the IRS-assisted reflected links. Therefore, the IRS-selection variable $\varphi_k \in \{0, 1, \dots, M\}$ is defined for every $k \in \mathcal{K}$ as

$$\varphi_k = \begin{cases} 0, & \text{user } k \text{ served by the direct link,} \\ m, & \text{user } k \text{ served by IRS } m. \end{cases} \quad (3)$$

B. Channel Model

Due to the long propagation distance between the satellite and ground nodes, the satellite-to-user (SU) and satellite-to-IRS (SR) links are dominated by large-scale path loss and rain attenuation following the ITU-R model [36]. The IRS-to-user (RU) link additionally exhibits small-scale Rayleigh fading and

distance-dependent path loss [4]. The true channel coefficients of the three links are modeled as

$$\begin{aligned}
 g_k^{SU} &= \frac{c\sqrt{G_S G_U \epsilon_k^{SU}}}{4\pi f_{SU} d_k^{SU}}, \\
 g_m^{SR} &= \frac{c\sqrt{G_S \epsilon_m^{SR}}}{4\pi f_{SR} d_m^{SR}}, \\
 g_{m,k}^{RU} &= \sqrt{G_U \cdot \Omega_{sf} \cdot \left(\frac{d_0}{d_{m,k}^{RU}}\right)^\alpha} \cdot \epsilon_{m,k}^{RU} \tilde{g}_{m,k}^{sf},
 \end{aligned} \tag{4}$$

for every $m \in \mathcal{M}$ and $k \in \mathcal{K}$, where c is the speed of light; G_S and G_U are the satellite transmit and user receive antenna gains; f_{SU} and f_{SR} are the corresponding carrier frequencies; d_k^{SU} , d_m^{SR} , and $d_{m,k}^{RU}$ denote the satellite-to-user, satellite-to-IRS, and IRS-to-user link distances; ϵ_k^{SU} , ϵ_m^{SR} , and $\epsilon_{m,k}^{RU}$ are the rain-attenuation coefficients modeled as lognormal random variables per the ITU-R standard; Ω_{sf} is the average shadow-fading power gain of the IRS-to-user link; α is the path-loss exponent; d_0 is the reference distance; and $\tilde{g}_{m,k}^{sf} \sim \mathcal{CN}(0, 1)$ represents the small-scale Rayleigh fading component. When the satellite-to-user line of sight is intersected by a building, the affected coefficient g_k^{SU} is scaled by a blocking factor $\beta_{\text{blk}} \in (0, 1]$, so that a blocked user retains a weak direct path rather than losing it entirely. Each channel coefficient is complex, with a random propagation phase uniformly distributed on $[0, 2\pi)$, which is tracked via Doppler compensation at the satellite.

Since the satellite acquires only an imperfect estimate of each channel coefficient, the estimated channel $\hat{g}$ and the true channel g are related by a multiplicative error model:

$$\hat{g} = g + \Delta g, \quad \Delta g = \kappa |g| \varepsilon, \quad \varepsilon \sim \mathcal{CN}(0, 1), \tag{5}$$

where $\kappa \geq 0$ is the relative error coefficient applied uniformly to all three links. This yields the estimated channels $\hat{g}_k^{SU}$, $\hat{g}_m^{SR}$, and $\hat{g}_{m,k}^{RU}$, which are used for all precoding and resource allocation design. The policy is designed on the estimated channels alone, the assignment actor consuming per-user path qualities and the phase actor the estimated cascaded coefficients, whereas the environment scores the achieved rate on the true channels g .

C. Transmission and Decoding

Following the RSMA principle [7], each user message s_k of user $k \in \mathcal{K}_g$ is split into a common sub-message $s_{c,g}$ and a private sub-message $s_{k,p,g}$. However, not all IRSs are necessarily selected by the agent at each decision step. Therefore, the number of active groups is denoted by G , where $G \leq M + 1$. We define the set of scheduled user messages after splitting as

$$\tilde{\mathbf{s}} = [s_{c,1}, s_{c,2}, \dots, s_{c,G}, \{s_{k,p,g}\}_{g \in \mathcal{G}, k \in \mathcal{K}_g}].$$

A single-antenna satellite precludes spatial beamforming, so each stream carries a scalar coefficient, $w_{c,g} \in \mathbb{C}$ for the common stream of group g and $w_{k,p,g} \in \mathbb{C}$ for the private stream of user $k \in \mathcal{K}_g$, and the transmitted signal is

$$x = \sum_{g \in \mathcal{G}} w_{c,g} s_{c,g} + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} w_{k,p,g} s_{k,p,g}, \quad (6)$$

where $s_{c,g}$ is the common stream intended for all users in group $\mathcal{K}_g$, and $s_{k,p,g}$ is the private stream intended only for user k in group g . The scalar precoding coefficients are constrained by the maximum transmit power of the satellite, i.e., $\sum_{g \in \mathcal{G}} |w_{c,g}|^2 + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} |w_{k,p,g}|^2 \leq P_S$, where P_S denotes the maximum transmit power of the satellite.

After receiving y_k , each user decodes its group-specific common stream and then its private stream by successive interference cancellation, following the standard RSMA decoding order [5], [7]. For user $k \in \mathcal{K}_g$, the SINRs for decoding the common stream $s_{c,g}$ and the private stream $s_{k,p,g}$ are respectively given by

$$\begin{aligned} \gamma_{k,c,g} &= \frac{|h_k w_{c,g}|^2}{\sum_{g' \neq g} |h_k w_{c,g'}|^2 + \sum_{g' \in \mathcal{G}} \sum_{j \in \mathcal{K}_{g'}} |h_k w_{j,p,g'}|^2 + \sigma^2}, \\ \gamma_{k,p,g} &= \frac{|h_k w_{k,p,g}|^2}{\sum_{g' \in \mathcal{G}} \sum_{\substack{j \in \mathcal{K}_{g'} \\ j \neq k}} |h_k w_{j,p,g'}|^2 + \sum_{g' \neq g} |h_k w_{c,g'}|^2 + \sigma^2}. \end{aligned} \quad (7)$$

Since the common stream $s_{c,g}$ is intended for all users in group $\mathcal{K}_g$, it must be successfully decoded by every user in that group. Therefore, the achievable common rate of group g is limited by the user with the lowest common-stream decoding rate. For user $k \in \mathcal{K}_g$, the common-stream decoding rate and the

resulting group common rate are given by

$$R_{k,c,g} = \log_2(1 + \gamma_{k,c,g}), \quad R_{c,g} = \min_{k \in \mathcal{K}_g} R_{k,c,g}. \quad (8)$$

Let $C_{k,g}$ denote the portion of the group common rate allocated to user $k \in \mathcal{K}_g$. The common-rate allocation must satisfy

$$\sum_{k \in \mathcal{K}_g} C_{k,g} = R_{c,g}, \quad C_{k,g} \geq 0. \quad (9)$$

After decoding and removing the common stream, user k decodes its private stream, achieving rate $R_{k,p,g} = \log_2(1 + \gamma_{k,p,g})$. The total achievable rate of user k is thus $R_k = R_{k,p,g} + C_{k,g}$, and the system sum-rate is

$$R_{\text{tot}} = \sum_{k \in \mathcal{K}} R_k = \sum_{g \in \mathcal{G}} R_{c,g} + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} R_{k,p,g}. \quad (10)$$

D. Optimization Problem

The objective is to maximize the system sum-rate by jointly optimizing the IRS-selection variables, the scalar precoding coefficients, the IRS phase-shift matrices, and the common-rate allocation. Specifically, the optimization variables are given by $\boldsymbol{\varphi} = \{\varphi_k\}_{k \in \mathcal{K}}$, $\mathbf{w} = \{w_{c,g}, w_{k,p,g}\}_{g \in \mathcal{G}, k \in \mathcal{K}_g}$, $\boldsymbol{\Phi} = \{\boldsymbol{\Phi}_m\}_{m \in \mathcal{M}}$, and $\mathbf{C} = \{C_{k,g}\}_{k \in \mathcal{K}_g, g \in \mathcal{G}}$. The sum-rate maximization problem is formulated as

$$\text{(P0): } \max_{\boldsymbol{\varphi}, \mathbf{w}, \boldsymbol{\Phi}, \mathbf{C}} R_{\text{tot}} \quad (11a)$$

$$\text{s.t. } \varphi_k \in \{0, 1, \dots, M\}, \quad \forall k \in \mathcal{K}, \quad (11b)$$

$$\sum_{g \in \mathcal{G}} |w_{c,g}|^2 + \sum_{g \in \mathcal{G}} \sum_{k \in \mathcal{K}_g} |w_{k,p,g}|^2 \leq P_S, \quad (11c)$$

$$\theta_{m,n} \in \mathcal{Q}_\theta, \quad \forall m \in \mathcal{M}, \forall n \in \mathcal{N}, \quad (11d)$$

$$\boldsymbol{\Phi}_m = \text{diag}(e^{j\theta_{m,1}}, \dots, e^{j\theta_{m,N}}), \quad \forall m, \quad (11e)$$

$$\sum_{k \in \mathcal{K}_g} C_{k,g} = R_{c,g}, \quad \forall g \in \mathcal{G}, \quad (11f)$$

$$C_{k,g} \geq 0, \quad \forall k \in \mathcal{K}_g, \forall g \in \mathcal{G}, \quad (11g)$$

$$\Pr\{R_k \geq R_k^{\min} \mid \hat{g}\} \geq 1 - \epsilon_o, \quad \forall k. \quad (11h)$$

In the above problem (P0), (11b) limits the values of the IRS-selection variables, which determine whether each user is served through the direct satellite-to-user link or one of the available IRS-assisted links. This choice also induces the grouping, so that $\mathcal{K}_g$ and G in the remaining constraints are themselves functions of φ . (11c) guarantees that the total transmit power of the satellite does not exceed the maximum power budget. (11d) restricts the phase shift of each IRS reflecting element to the discrete quantized phase-angle set, while constraint (11e) defines the corresponding passive IRS phase-shift matrix. Constraint (11f) requires that the allocated common rates fully exhaust the group common rate, i.e., the entire decodable common rate of each group is distributed among its members with no waste. Constraint (11g) ensures non-negativity of the allocated common rates, and constraint (11h) [37], [38] imposes the per-user minimum-rate requirement at an outage level ϵ_o .

IV. PROPOSED HQC-HAC FRAMEWORK

A. Overall Framework

Problem (P0) is a mixed-integer non-convex program. The assignment φ is discrete over a set of size $(M + 1)^K$, the per-element phase shifts are confined to the quantized set $\mathcal{Q}_\theta$, and both couple to the continuous powers $\mathbf{w}$ and common-rate shares $\mathbf{C}$ through the SINR expressions, which also render R_{tot} non-concave. The problem is therefore NP-hard in general [10], and the model-based relaxations discussed in Section II, such as semidefinite relaxation [11] and alternating optimization, carry the limitations set out there. We instead learn a policy that maps the observed state directly to the joint action.

The four decision variables exhibit a strict sequential dependency: the assignment φ fixes which IRS panels are active, and thereby the number of active groups G and the dimension of the common-power vector $\mathbf{w}_c$; the active set governs the effective channels that underpin the phase-shift design Φ ; the resulting channel gains shape the power-allocation landscape $(\mathbf{w}_p, \mathbf{w}_c)$; and the allocated powers delimit the feasible common-rate split $\mathbf{C}$. We adopt a hierarchical actor–critic decomposition that mirrors this ordering: a high-level actor commits to the discrete assignment, and three low-level actors resolve the continuous variables in sequence, each conditioned on the decisions made upstream.

The high-level assignment remains the principal source of difficulty, since $\{0, 1, \dots, M\}^K$ is non-differentiable and grows exponentially in both K and M . We therefore place a variational quantum circuit (VQC) at the top of the hierarchy, where an n_q -qubit state parameterizes a distribution over

this space compactly. Once φ is fixed the remaining sub-problems are smooth, continuous, and low-dimensional, and are handled by classical multi-layer perceptrons (MLPs); reserving the quantum circuit for the combinatorial bottleneck is what renders the framework *hybrid* quantum–classical.

At each decision step t the raw channel observation is condensed into the affinity state $\mathbf{s}_t$, which both the quantum actor and the shared critic consume, and an autoencoder further compresses it to the latent $\mathbf{z}_t \in \mathbb{R}^{2n_q}$ that the n_q -qubit register encodes (Section IV-B). Following the data flow in Fig. 2, the VQC maps $\mathbf{z}_t$ to a measurement vector, which the quantum actor’s policy layer converts into a categorical distribution over candidate assignments; sampling it yields $\boldsymbol{\varphi}$. The three sub-actors then act in sequence: the Phase MLP produces the quantized phase indices $\boldsymbol{\theta}_{\text{idx}}$ and hence $\boldsymbol{\Phi}$, the Power MLP allocates the private and common powers $(\mathbf{w}_p, \mathbf{w}_c)$, and the Common-Rate MLP distributes the group common rates $\mathbf{C}$

among users. The joint action is applied to the environment, which returns the next state and the reward of (22).

All four actors and the shared critic are trained together by proximal policy optimization (PPO) [39] with entropy regularization annealed over training. Sections IV-B–IV-E describe the autoencoder, the quantum circuit, and the training procedure in detail.

B. Autoencoder-Based Latent State Representation

Directly encoding the raw environment observation into a quantum register is impractical, since its dimension $M(3 + 2K + 2N) + 3K$ grows with the number of IRS elements N , which is unrelated to the IRS-user assignment decision. The quantum actor instead operates on the compact *affinity state* $\mathbf{s}_t \in \mathbb{R}^{K(M+2)+2M}$ of Section IV, which retains only the information pertinent to link assignment. Even so, $\mathbf{s}_t$ still exceeds the register, requiring one qubit per two components. The compression is therefore marginal at the smallest configuration, yet it is precisely what renders the larger ones representable. The affinity state is structured as

$$\mathbf{s}_t = [a_{k,m}, R_k^{\min}, |\hat{g}_k^{SU}|, q_m, p_m], \quad (12)$$

where $a_{k,m} = |\hat{g}_m^{SR}| \sum_{n=1}^N |\hat{g}_{m,n,k}^{RU}|$ is the IRS path quality for user k via IRS m , $q_m = \frac{1}{K} \sum_{k=1}^K a_{k,m}$ is the mean affinity of IRS m across all users, and $p_m = |\hat{g}_m^{SR}|^2$ is the received satellite-IRS power. The affinity, direct-link, and IRS blocks are standardized separately before encoding, yielding $\bar{\mathbf{s}}_t$.

A dual-branch encoder maps $\bar{\mathbf{s}}_t$ to the latent vector $\mathbf{z}_t \in \mathbb{R}^{2n_q}$, where n_q is the number of qubits. As illustrated in Fig. 3, the encoder comprises two parallel branches that process IRS-level and user-level information separately prior to concatenation. The *IRS branch* maps the IRS summary statistics $[q_m, p_m]_{m=1}^M \in \mathbb{R}^{2M}$ through a two-layer MLP,

$$\mathbf{z}^{\mathcal{R}} = \mathbf{W}_2^{\mathcal{R}} \text{ReLU}(\mathbf{W}_1^{\mathcal{R}}[q_1, p_1, \dots, q_M, p_M]^\top + \mathbf{b}_1^{\mathcal{R}}) + \mathbf{b}_2^{\mathcal{R}} \in \mathbb{R}^{2M}, \quad (13)$$

where $\mathbf{W}_1^{\mathcal{R}} \in \mathbb{R}^{16 \times 2M}$ and $\mathbf{W}_2^{\mathcal{R}} \in \mathbb{R}^{2M \times 16}$. The *user branch* applies a shared two-layer micro-encoder independently to each user k , following the parameter-sharing scheme of set-encoding architectures [40],

$$\mathbf{z}_k = \mathbf{W}_2^{\mathcal{U}} \text{ReLU}(\mathbf{W}_1^{\mathcal{U}}[a_{k,1}, \dots, a_{k,M}, R_k^{\min}, |\hat{g}_k^{SU}|]^\top + \mathbf{b}_1^{\mathcal{U}}) + \mathbf{b}_2^{\mathcal{U}} \in \mathbb{R}^2, \quad \forall k \in \mathcal{K}, \quad (14)$$

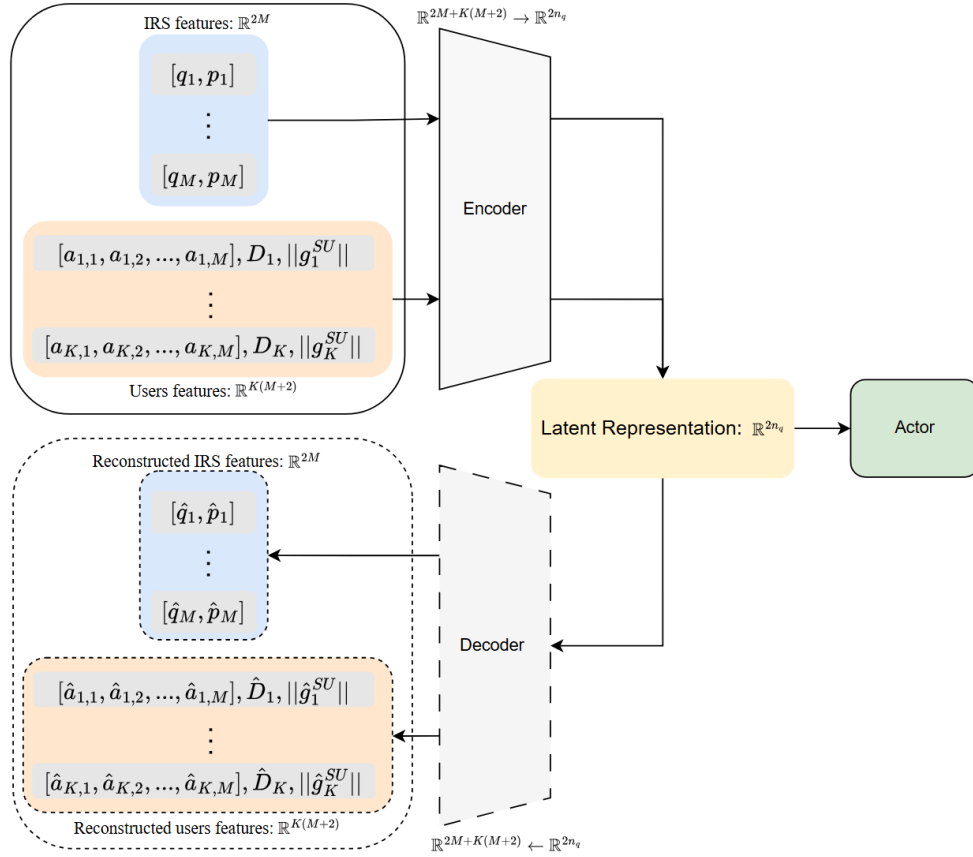


Fig. 3: Dual-branch autoencoder architecture for latent state compression. The encoder (top path) processes IRS summary features $[q_m, p_m]$ and per-user affinity features $[a_{k,m}, R_k^{\min}, |\hat{g}_k^{SU}|]$ through separate branches before concatenating into the latent representation $\mathbf{z}_t \in \mathbb{R}^{2n_q}$, which is forwarded to the quantum actor. The decoder (bottom path, active during training only) reconstructs the input feature vector $\hat{\mathbf{s}}_t$ from $\mathbf{z}_t$ and evaluates the reconstruction loss $\mathcal{L}_{AE} = \frac{1}{2} \|\hat{\mathbf{s}}_t - \bar{\mathbf{s}}_t\|^2$ to regularize the latent space.

where $\mathbf{W}_1^{\mathcal{U}} \in \mathbb{R}^{8 \times (M+2)}$ and $\mathbf{W}_2^{\mathcal{U}} \in \mathbb{R}^{2 \times 8}$ are shared across all K users. A linear map $\mathbf{W}_{\text{proj}} \in \mathbb{R}^{2(n_q - M) \times 2K}$ projects the K stacked embeddings onto the user block, $\mathbf{z}^{\mathcal{U}} = \mathbf{W}_{\text{proj}}[\mathbf{z}_1^\top, \dots, \mathbf{z}_K^\top]^\top$, and the latent concatenates the two branches as $\mathbf{z}_t = [\mathbf{z}^{\mathcal{U}} \parallel \mathbf{z}^{\mathcal{R}}] \in \mathbb{R}^{2n_q}$, whose blocks are loaded onto the user qubits $\mathcal{U}$ and the IRS qubits $\mathcal{R}$ of Section IV-C. Keeping the branches apart is what gives that mapping content. Both the cross-block bridges and the user-IRS observables of Section IV-C are defined over this partition, so a single encoder that mixed the two sides would leave a user qubit and an IRS qubit holding arbitrary combinations, and observables meant to track the assignment would reduce to a generic set. The separation also leaves the entangling gates as the only stage at which features from the two blocks are multiplied together.

The decoder is a single stack rather than a mirror of the two-branch encoder. It expands $\mathbf{z}_t$ through rectified-linear hidden layers, after which a linear output layer returns the reconstruction $\hat{\mathbf{s}}_t$ to the width

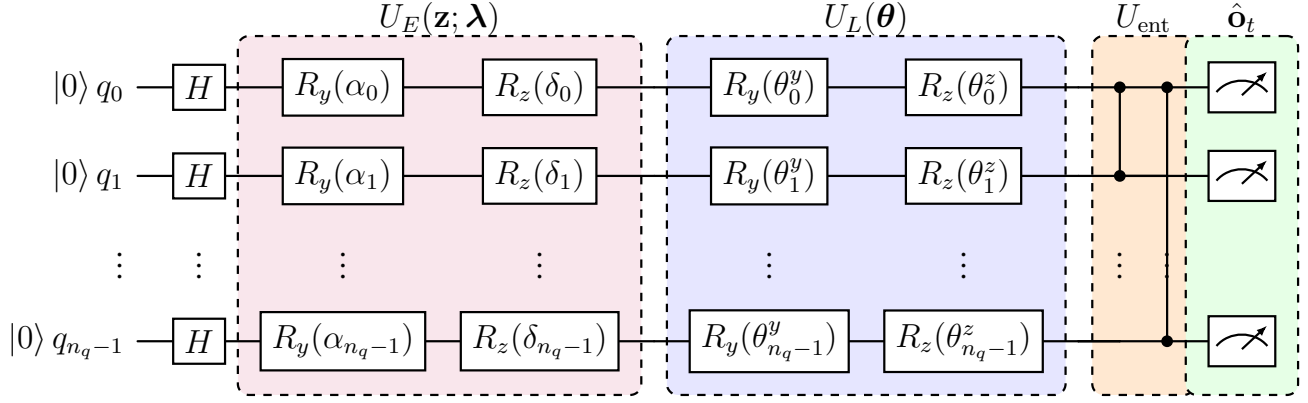


Fig. 4: Variational quantum circuit (VQC) of the quantum actor, shown for representative qubits $q_0, q_1, \dots, q_{n_q-1}$. Following $H^{\otimes n_q}$, each layer applies the encoding block $U_E(\mathbf{z}; \boldsymbol{\lambda})$, variational block $U_L(\boldsymbol{\theta})$, and entangling block U_{ent} , with data re-uploading across layers. The measurement block marks the readout stage, whose observable set is defined in (19).

of $\mathbf{s}_t$, producing the reconstruction loss

$$\mathcal{L}_{AE} = \frac{1}{2} \|\hat{\mathbf{s}}_t - \bar{\mathbf{s}}_t\|^2, \quad (15)$$

where $\bar{\mathbf{s}}_t$ is the standardized affinity state. The decoder is used only during training and is discarded at inference, where only the encoder and the latent $\mathbf{z}_t$ are forwarded to the VQC.

The autoencoder is trained jointly and end-to-end with the rest of the framework, following the latent-observation approach of [20]. The reconstruction loss in (15) enters the policy objective as an auxiliary regularizer of weight α_{AE} , so the encoder receives gradients from both $\mathcal{L}_{AE}$ and the reinforcement-learning loss back-propagated through the quantum actor. The reconstruction term is retained throughout as a structural regularizer that preserves the informativeness of $\mathbf{z}_t$, with α_{AE} balancing representational fidelity against task performance.

Joint optimization is preceded by a short reconstruction-only *warm-up* in which the autoencoder alone is updated and every other component is held fixed. It is fitted on synthetic samples from the standardized input distribution rather than on environment rollouts, which suffices to condition the encoder so that the first policy gradients do not act upon an untrained representation.

C. Variational Quantum Circuit

The latent vector $\mathbf{z}_t \in \mathbb{R}^{2n_q}$ from the dual-branch encoder is processed by a variational quantum circuit (VQC) acting on an n_q -qubit register, which maps it to a compact feature vector $\hat{\mathbf{o}}_t$. The dimension $2n_q$

supplies one rotation angle per qubit along each of two encoding axes (defined below). Preparing an n_q -qubit state spans a 2^{n_q} -dimensional Hilbert space, so the circuit parameterizes a distribution over the assignment space with a compact trainable-parameter budget (Section V-B).

1) *Data Encoding*: Encoding begins with a layer normalization [41] of the latent vector,

$$\tilde{\mathbf{z}}_t = \gamma \odot \frac{\mathbf{z}_t - \mu_z}{\sigma_z + \epsilon_{\text{LN}}} + \mathbf{b}_{\text{LN}}, \quad (16)$$

where μ_z and σ_z are the mean and standard deviation of $\mathbf{z}_t$, and $\gamma, \mathbf{b}_{\text{LN}} \in \mathbb{R}^{2n_q}$ are a learnable affine trained jointly with the quantum parameters. The affine restores the DC component that a plain zero-mean normalization removes, which would otherwise cancel part of the gradient reaching the encoding scales. The normalized vector is then mapped to per-qubit rotation angles through trainable scales $\lambda_y, \lambda_z \in \mathbb{R}^{n_q}$,

$$\alpha_i = \pi \tanh(\lambda_i^y \tilde{z}_{2i}), \quad \delta_i = \pi \tanh(\lambda_i^z \tilde{z}_{2i+1}), \quad i = 0, \dots, n_q - 1, \quad (17)$$

so that consecutive latent components are paired onto a common qubit. The pair (α_i, δ_i) places one latent component on each and n_q qubits accommodate the full $2n_q$ -dimensional latent vector. The $\tanh$ confines every angle to $(-\pi, \pi)$ and rules out over-rotation, while λ lets the circuit amplify or attenuate the influence of individual latent dimensions. The scales are clipped to $|\lambda| \leq \lambda_{\text{max}} = 1.5$ after each update, since $\tanh$ saturates beyond this range and the gradient reaching the encoder would vanish with it.

2) *Circuit Ansatz*: After a Hadamard layer places the register in a uniform superposition, L identical layers apply the encoding block U_E , the variational block U_L , and the entangling block U_{ent} in sequence, as Fig. 4 illustrates. The state prepared at step t is therefore

$$|\psi_t\rangle = \left[\prod_{\ell=L}^1 U_{\text{ent}} U_L(\boldsymbol{\theta}_\ell) U_E(\tilde{\mathbf{z}}_t; \boldsymbol{\lambda}) \right] H^{\otimes n_q} |0\rangle^{\otimes n_q}, \quad (18)$$

where ℓ indexes the L layers. *Data re-uploading* [42] inserts U_E within every layer, which enriches the class of functions the circuit can represent. A model of this form is a partial Fourier series in its input, whose accessible frequency spectrum is set by the encoding gates and broadens with each repetition, while the variational and entangling blocks govern the coefficients [43]. The entangling block carries no parameters, so the circuit is trained through two sets of angles, the variational angles $\boldsymbol{\theta}_\ell$ and the encoding angles, the latter being trainable through the scales $\boldsymbol{\lambda}$ of (17).

The nearest-neighbor CZ chain couples each qubit to its successor, generating correlations beyond those attainable by a product state. Since the latent vector is partitioned into a user segment and an IRS segment, the register likewise separates into a user block and an IRS block, which the chain links only at their boundary. A further M CZ bridges connect IRS qubits to user qubits distributed across the block, so that the circuit can correlate a user qubit with an IRS qubit to which that user may be assigned. The ansatz draws exclusively on H , R_y , R_z , and CZ gates, placing it within the hardware-efficient family that near-term devices implement natively [44].

3) *Measurement*: The circuit terminates in the measurement stage of Fig. 4, the only point at which the register yields a quantity a classical module can consume. Whatever the ansatz has encoded reaches the policy solely through the expectation values taken there, so the choice of observables fixes the interface between the two halves of the actor. The conventional choice is a single Pauli- Z per qubit, which returns n_q numbers and describes each qubit in isolation. The set adopted here departs from that convention, organizing the observables around the structure of the assignment and admitting two-qubit terms, with the intent of presenting ξ a richer and better aligned summary than n_q independent marginals. The readout, whose entries are expectations in the state $|\psi_t\rangle$ of (18), is defined as

$$\hat{\mathbf{o}}_t = [\langle Z_u \rangle, \langle Z_u Z_r \rangle, \frac{1}{|\mathcal{U}|-1} \sum_{u' \neq u} \langle Z_u Z_{u'} \rangle, \langle Z_r \rangle] \in [-1, 1]^{N_Q}, \quad (19)$$

where $\mathcal{U} = \{0, \dots, n_q - M - 1\}$ and $\mathcal{R} = \{n_q - M, \dots, n_q - 1\}$ index the user and IRS qubits. The four blocks report the marginal state of each user qubit, its correlation with every IRS qubit, its mean correlation with the remaining user qubits, and the marginal state of each IRS qubit, so that every entry is indexed by the objects on which the assignment acts.

Every observable in (19) acts on at most two qubits, so the readout is local; together with the narrow and shallow ansatz, this situates the model within the local-cost regime shown to remain trainable at shallow depth [45] rather than the deep, globally measured regime in which gradients vanish exponentially in the qubit count [46]. The two-qubit entries of (19) differ from the products of their one-qubit counterparts only insofar as the entangling block correlates the two registers, so whatever they contribute originates in the ansatz rather than in the readout. Whatever the observable set omits of the prepared state is not recoverable downstream, but the skip connection of Section IV-C4 routes $\mathbf{z}_t$ past the measurement, so

a lossy readout nevertheless cannot leave the policy worse off than a classical layer reading the latent vector alone. The observables mutually commute, hence a single measurement basis suffices for the entire vector and one set of shots estimates all N_Q entries. Each entry is obtained by averaging over repeated measurements rather than inferred from a single outcome, which keeps the distribution the behavior policy samples from close to the analytic one underlying both log-probabilities in the PPO ratio. The behavior policy draws on $S_{\text{tr}} = 1500$ shots per estimate during training and $S_{\text{ev}} = 3000$ at inference. By contrast, the learning signal is evaluated on the exact statevector of the simulated circuit, so sampling noise perturbs exploration and evaluation while leaving the gradient unaffected.

4) *Softmax Policy Layer*: Recovering a concrete assignment from the circuit requires a distribution over the available actions, since the measurement returns real-valued expectations rather than a decision. Jerbi *et al.* [17] obtain one by passing a weighted combination of the measured observables through a softmax, the construction they term a softmax-PQC and adopt as the policy parameterization of a parametrized quantum agent. The layer used here follows that construction, and producing an assignment from it asks two things, that the readout be mapped onto the shape the decision takes and that the user–IRS structure the circuit forms be available to that mapping.

The first is dimensional, since the register fixes the readout at N_Q entries whereas the assignment is specified by $K(M+1)$ logits, and the layer therefore performs the projection between the two alongside the decision itself. The second concerns what $\hat{\mathbf{o}}_t$ carries, and follows from how $\mathbf{z}_t$ is built. The dual-branch encoder forms the user and IRS blocks separately and never multiplies them, while the entangling gates couple the two registers and the user–IRS observables of (19) report that coupling, so $\hat{\mathbf{o}}_t$ supplies exactly the products across the two sides that $\mathbf{z}_t$ holds only separately. Concatenating them, $\mathbf{h}_t = [\mathbf{z}_t \parallel \hat{\mathbf{o}}_t] \in \mathbb{R}^{2n_q + N_Q}$, presents the layer with the per-side features and those cross terms at once, neither of which a linear map could assemble from the other. A single linear map followed by a trainable inverse temperature τ then yields a per-user assignment distribution,

$$\boldsymbol{\pi}_k = \text{softmax}\left(\tau [\mathbf{W}\mathbf{h}_t + \mathbf{b}]_{k(M+1):(k+1)(M+1)}\right) \in \Delta^M, \quad k \in \mathcal{K}, \quad (20)$$

where $\mathbf{W} \in \mathbb{R}^{(2n_q + N_Q) \times K(M+1)}$ and $\mathbf{b} \in \mathbb{R}^{K(M+1)}$ are the weight matrix and bias of the layer, the slice extracts user k 's $M+1$ logits from their shared output, and Δ^M is the probability simplex over the direct

link and the M panels. The inverse temperature τ is a trained scalar applied to every logit, sharpening π_k as it grows and flattening it as it falls, so the policy governs its own balance between committing to the leading choice and continuing to explore. The assignment is sampled $\varphi_k \sim \pi_k$ during training and taken greedily, $\varphi_k = \arg \max \pi_k$, at evaluation.

Softmaxing the blocks independently leaves the K assignments conditionally independent given $\mathbf{h}_t$, factorizing the $(M+1)^K$ action space into K categoricals and rendering the log-probability $\sum_k \log \pi_k(\varphi_k)$ and the entropy separable across users. At $(2n_q + N_Q) \times K(M+1)$ weights the layer remains small, and Table VI itemizes it against the classical assignment actor. The variational angles $\theta_1, \dots, \theta_L$, the encoding scales λ_y, λ_z and the affine transformation of (16) are updated by gradient descent alongside ξ ; they differ only in that the Jacobian of $\hat{\mathbf{o}}_t$ with respect to them is estimated by SPSA(Section IV-E).

D. Classical Sub-Actors

Once the assignment φ is drawn, the three remaining decision variables are produced in sequence by small multilayer perceptrons conditioned on φ and on the latent $\mathbf{z}_t$. Each satisfies its own constraint through the form of its output, so no infeasible action is sampled. The phase decision is discrete and is emitted as a categorical distribution, whereas the power and common-rate decisions are points on a simplex and are drawn from Dirichlet distributions, which is what allows the sampled vector to be the executed one. Executing a deterministic normalization while scoring a separately drawn index would leave the importance ratio of (24) comparing two different actions.

The *Phase MLP* selects a quantization level for every reflecting element. Its input is assembled per element, row (m, n) carrying the unit phasors of the cascade channel $c_{m,n,k} = (\hat{g}_m^{SR})^* \hat{g}_{m,n,k}^{RU}$ of the users routed to IRS m , the unit phasors of the direct links $\hat{g}_k^{SU}$, the latent $\mathbf{z}_t$, and the routing mask. Magnitudes are divided out because only angles matter for alignment, and the direct link is kept as the reference the elements align against, $\theta_{m,n} \approx \arg \hat{g}_k^{SU} - \arg c_{m,n,k}$; without it the power-maximizing phase is not a function of the state at all. One weight set is shared across every element and surface and applied to them as a batch, so the module emits one logit per quantization level per element and its parameter count does not depend on N .

The *Power MLP* factors the budget into three independent Dirichlet heads, a two-way split between the common and private shares, an allocation of the common share over the direct group and the M

Algorithm 1 HQC-HAC training

Require: $\Theta = \{\omega, \eta, \xi, \Theta_\Phi, \Theta_w, \Theta_C\}$, ψ , $E=12$, $T=200$, $N_{\text{ep}}=6$

```

1:  $\omega \leftarrow \arg \min \mathcal{L}_{AE}$   $\triangleright \Theta \setminus \omega, \psi$  frozen
2:  $\eta, \xi \leftarrow \arg \min \mathbb{E}[-\log \pi_\varphi(\varphi^*)]$ 
3:  $\Theta_\Phi \leftarrow \arg \min \mathbb{E}[-\log \pi_\Phi(\Phi^* | \varphi^*)]$ 
4: repeat
5:    $\mathcal{B} \leftarrow \emptyset$ 
6:   for  $e = 1 : E$  do
7:     for  $t = 1 : T$  do
8:        $\mathbf{s}_t \leftarrow \mathbf{s}(\hat{g}_t)$  (12),  $\mathbf{z}_t \leftarrow \text{AE}_\omega(\mathbf{s}_t)$ 
9:        $\varphi_t \sim \pi_\varphi(\cdot | \mathbf{s}_t)$  (20)
10:       $\Phi_t \sim \pi_\Phi(\cdot | \hat{g}_t, \mathbf{z}_t, \varphi_t)$ 
11:       $\mathbf{w}_t \sim \pi_w(\cdot | \hat{g}_t, \mathbf{z}_t, \varphi_t, \Phi_t)$ 
12:       $\mathbf{C}_t \sim \pi_C(\cdot | \varphi_t, \Phi_t, \mathbf{w}_t)$ 
13:       $r_t \leftarrow r(\mathbf{a}_t; g_t)$  (22),  $\mathbf{s}_{t+1} \sim \mathcal{P}(\cdot | \mathbf{s}_t, \mathbf{a}_t)$ 
14:       $\mathcal{B} \leftarrow \mathcal{B} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r_t, V_t, \{\log \pi_j^{\text{old}}\}_j)\}$ 
15:    end for
16:  end for
17:   $\hat{A} \leftarrow \text{GAE}(\mathcal{B})$  (26),  $\hat{A} \leftarrow (\hat{A} - \mu_{\hat{A}})/\sigma_{\hat{A}}$ 
18:  for  $n = 1 : N_{\text{ep}}$  do
19:    for all  $\mathcal{B}_m \subset \mathcal{B}$  do
20:       $\hat{\mathbf{J}} \leftarrow \text{SPSA}(\partial \hat{\mathbf{o}}_t / \partial \eta)$  (28)
21:       $\nabla_\eta \mathcal{J}_\varphi \leftarrow \hat{\mathbf{J}}^\top \nabla_{\hat{\mathbf{o}}_t} \mathcal{J}_\varphi$ 
22:       $\Theta \leftarrow \Theta + \alpha \nabla_\Theta \mathcal{J}$  (25)
23:       $\psi \leftarrow \psi - \alpha_\psi \nabla_\psi \mathcal{L}_V$  (27)
24:    end for
25:  end for
26:   $\beta \leftarrow \max(\beta_{\min}, \beta - \Delta\beta)$ ,  $\alpha \leftarrow \text{decay}(\alpha)$ ,  $(\mu, \sigma) \leftarrow \text{PopArt}(G)$ 
27: until episode budget exhausted

```

IRS groups with inactive surfaces masked, and an allocation of the private share over the K users. The budget $\sum_g w_c^g + \sum_k w_{p,k} = P_S$ then holds identically, and the factorization gives each axis its own gradient in place of one softmax over mixed slots. The three factors are treated as independent, $\log \pi_w = \log \pi_{\text{split}} + \log \pi_{\text{common}} + \log \pi_{\text{private}}$.

The *Common-Rate MLP* reads the demands $R_k^{\min}$, the private rates already achieved $R_{k,p,g}$, the common rate available to each user's group $R_{c,g}$, the resulting shortfalls ς_k , and the assignment φ_k , and draws, for each group, a Dirichlet vector that apportions the common rate among its members, so $\sum_{k \in g} C_k \leq R_{c,g}$ is satisfied by construction.

All three consume the shared advantage of Section IV-E and are updated by the same clipped surrogate as the quantum actor.

E. Entropy-Regularized Hierarchical Actor-Critic Training

The four actors of Section IV are trained together with a single shared critic in an entropy-regularized actor-critic scheme optimized by proximal policy optimization (PPO) [39]. The quantum actor and the three classical sub-actors constitute a *joint hierarchical policy* that factorizes the composite action, while the critic supplies a common advantage signal that couples their updates toward the shared goal of sum-rate maximization under per-user QoS constraints.

1) *Hierarchical Policy Update*: We cast the sequential resource-allocation task as a Markov decision process $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma_d)$, where $\gamma_d \in (0, 1)$ is the discount factor, written with a subscript to keep it apart from the SINR γ . The *state* $\mathbf{s}_t$ is the affinity feature vector of (12), built from the imperfect estimates $\hat{g}$, and the *action* is the joint decision $\mathbf{a}_t = (\boldsymbol{\varphi}, \boldsymbol{\Phi}, \mathbf{w}, \mathbf{C})$. The hierarchy of Sections IV-C and IV-D realizes this action as a chain,

$$\begin{aligned} \pi(\mathbf{a}_t \mid \mathbf{s}_t, \hat{g}) &= \pi_{\boldsymbol{\varphi}}(\boldsymbol{\varphi} \mid \mathbf{s}_t) \pi_{\boldsymbol{\Phi}}(\boldsymbol{\Phi} \mid \hat{g}, \mathbf{z}_t, \boldsymbol{\varphi}) \\ &\times \pi_{\mathbf{w}}(\mathbf{w} \mid \hat{g}, \mathbf{z}_t, \boldsymbol{\varphi}, \boldsymbol{\Phi}) \pi_{\mathbf{C}}(\mathbf{C} \mid \boldsymbol{\varphi}, \boldsymbol{\Phi}, \mathbf{w}), \end{aligned} \quad (21)$$

in which every factor is conditioned on the decision-time context and on all decisions taken upstream of it. The affinity state $\mathbf{s}_t$ is what the assignment actor and the critic consume, whereas the three sub-actors read the quantities of Section IV-D directly from $\hat{g}$, which $\mathbf{s}_t$ does not subsume since it carries no per-element index. This is what *hierarchical* means operationally here, and it gives $\log \pi = \sum_j \log \pi_j$, so the surrogate below applies to each actor separately. Once the action is applied, the environment advances the user positions and channels to yield the next state $\mathbf{s}_{t+1}$. The process has no absorbing state, an episode being a fixed-length truncation rather than a run to termination, so the bootstrap of (26) applies at every step.

The *reward* links the optimization objective (11a) to the QoS constraint (11h) through a soft quadratic penalty,

$$r_t = R_{\text{tot}} - \lambda_D \sum_{k \in \mathcal{K}} \left(\frac{\max(0, R_k^{\min} - R_k)}{R_k^{\min} + \epsilon_q} \right)^2, \quad (22)$$

where $R_k = R_{k,p,g} + C_{k,g}$ is the total achievable rate of user k , $\lambda_D > 0$ weights the penalty, and ϵ_q is a small constant that guards the denominator. The penalty vanishes when every user meets its demand and grows quadratically with the normalized shortfall otherwise, so maximizing r_t steers the policy toward the

constrained sum-rate optimum of (P0) without imposing a hard, non-differentiable feasibility boundary. The rates R_{tot} and R_k that enter (22) are those *achieved* on the true channel g , whereas every factor of (21) is conditioned on the estimate $\hat{g}$ alone.

A rollout stores, for every transition, the tuple $(\mathbf{s}_t, \mathbf{a}_t, r_t, V_t)$ together with the behavior log-probabilities $\log \pi^{j, \text{old}}$ of all four actors, which are what the importance ratios below are measured against. What the subsequent update maximizes is the entropy-regularized expected return,

$$\mathcal{J}(\Theta) = \mathbb{E} \left[\sum_t \gamma_d^t \left(r_t + \beta \sum_j H(\pi_j) \right) \right], \quad (23)$$

where $\Theta = \{\omega, \eta, \xi, \Theta_\Phi, \Theta_w, \Theta_C\}$ gathers the encoder, circuit and policy-layer parameters of the quantum actor together with those of the three sub-actors, $H(\pi_j) = -\sum \pi_j \log \pi_j$ is the entropy of the policy of actor j , and $\beta > 0$ weights an entropy bonus that sustains exploration over the combinatorial assignment space and prevents premature convergence to a deterministic policy.

Let $j \in \{\varphi, \Phi, w, C\}$ index the four actors and let $\rho_t^j = \pi_j(a_t^j | \cdot) / \pi^{j, \text{old}}(a_t^j | \cdot)$ denote the importance ratio between the current and behavior policies. Every actor is updated with the clipped PPO surrogate

$$\mathcal{L}_{\text{PPO}}^j = \mathbb{E} \left[\min(\rho_t^j \hat{A}_t, \text{clip}(\rho_t^j, 1 - \epsilon_c, 1 + \epsilon_c) \hat{A}_t) \right], \quad (24)$$

where ϵ_c is the clip range. Clipping bounds each update so that a large change in the high-level assignment cannot abruptly shift the conditioning of the downstream actors, and it permits several optimization epochs over the same batch. Across those epochs the upstream actions are replayed from the buffer rather than resampled, so the conditioning of each sub-actor in (21) is held fixed and its ratio compares two distributions over the same argument. Combining the clipped surrogate with the entropy bonus, the supervised term of the oracle teacher and, for the quantum actor, the autoencoder reconstruction loss (15), the per-actor training objectives become

$$\begin{aligned} \mathcal{J}_\varphi &= \mathcal{L}_{\text{PPO}}^\varphi + \beta H(\pi_\varphi) - w_\varphi \mathcal{L}_{\text{sup}}^\varphi - \alpha_{AE} \mathcal{L}_{AE}, \\ \mathcal{J}_j &= \mathcal{L}_{\text{PPO}}^j + \beta H(\pi_j) - w_j \mathcal{L}_{\text{sup}}^j, \quad j \in \{\Phi, C\}, \\ \mathcal{J}_w &= \mathcal{L}_{\text{PPO}}^w + \beta H(\pi_w), \end{aligned} \quad (25)$$

where $\mathcal{L}_{\text{sup}}^j = -\log \pi_j(a_j^*)$ is the supervised term of the oracle teacher described below, evaluated at a

reference action a_j^* and weighted by w_j . The power actor carries no teacher, its counterpart being the fairness blend of the same subsection. The advantage $\hat{A}_t$ is common to all four actors and is supplied by the critic described next.

2) *Shared Critic*: Every actor updates against the same estimate of how advantageous its action was relative to the current policy, and one shared critic supplies that estimate for the whole hierarchy. The critic $V(\mathbf{s}_t; \boldsymbol{\psi})$ is an MLP that estimates the state value and acts as a variance-reducing baseline. The advantage shared by all actors is obtained by generalized advantage estimation [47],

$$\hat{A}_t = \sum_{l \geq 0} (\gamma_d \lambda_{\text{GAE}})^l \delta_{t+l}, \quad \delta_t = r_t + \gamma_d V(\mathbf{s}_{t+1}; \boldsymbol{\psi}) - V(\mathbf{s}_t; \boldsymbol{\psi}), \quad (26)$$

where $\lambda_{\text{GAE}} \in [0, 1]$ trades bias against variance. Conditioning the baseline on the state alone makes $\hat{A}_t$ a low-variance signal common to the whole hierarchy and avoids the gradient self-cancellation of a state-action baseline. Advantages are standardized across each rollout batch before use. The critic's parameters are fitted by regressing toward the resulting λ -return target $G_t = \hat{A}_t + V(\mathbf{s}_t; \boldsymbol{\psi})$,

$$\mathcal{L}_V(\boldsymbol{\psi}) = (G_t - V(\mathbf{s}_t; \boldsymbol{\psi}))^2, \quad (27)$$

with $V(\mathbf{s}_t; \boldsymbol{\psi})$ further stabilized by PopArt value-target normalization [48], under which the network regresses a running-mean/variance-normalized target while its output layer is rescaled at every statistics update to preserve the real-valued prediction, decoupling the learning rate from the reward scale, which otherwise drifts substantially as the policy improves. No separate target network or Polyak averaging is used; the multi-step GAE bootstrap already yields a sufficiently low-variance regression target given the entropy-regularized, clipped PPO updates.

3) *Training Schemes*: Beyond the clipped surrogate itself, the optimization is shaped by an estimator for the quantum gradient, a supervised warm-up, two annealing schedules, and two auxiliary terms.

The encoding scales, the affine transformation of (16) and the variational angles, collectively $\boldsymbol{\eta} = \{\boldsymbol{\lambda}_y, \boldsymbol{\lambda}_z, \boldsymbol{\gamma}, \mathbf{b}_{\text{LN}}, \boldsymbol{\theta}\}$, enter the policy only through the circuit. Their Jacobian $\partial \hat{\mathbf{o}}_t / \partial \boldsymbol{\eta}$ admits an exact evaluation by the parameter-shift rule [49], [50], which costs two circuit evaluations per parameter and therefore scales linearly with $\dim(\boldsymbol{\eta})$. We estimate it instead by simultaneous perturbation stochastic approximation (SPSA) [51], which perturbs every parameter at once along a Rademacher direction $\boldsymbol{\Delta}$ and

recovers each component from one pair of evaluations,

$$\hat{J}_{o,i} = \frac{1}{R} \sum_{r=1}^R \frac{\hat{o}(\boldsymbol{\eta} + c_s \boldsymbol{\Delta}^{(r)}) - \hat{o}(\boldsymbol{\eta} - c_s \boldsymbol{\Delta}^{(r)})}{2 c_s} \Delta_i^{(r)}, \quad \frac{\partial \mathcal{J}_\varphi}{\partial \eta_i} = \sum_o \frac{\partial \mathcal{J}_\varphi}{\partial \hat{o}_o} \hat{J}_{o,i}, \quad (28)$$

where the $\boldsymbol{\Delta}^{(r)}$ are independent Rademacher vectors, $c_s = 0.1$ is the perturbation step, and $R = 16$ repetitions average the estimate down. Since $\mathbb{E}[\Delta_i \Delta_j] = \delta_{ij}$, the construction is unbiased up to $O(c_s^2)$, and its cost is $2R$ circuit evaluations per sample irrespective of $\dim(\boldsymbol{\eta})$, against $2 \dim(\boldsymbol{\eta})$ for the parameter-shift rule. The factor $\partial \mathcal{J}_\varphi / \partial \hat{o}_t$ multiplying the estimated Jacobian is itself exact, as are the gradients of ξ , of the entropy term and of the autoencoder reconstruction. This efficiency is obtained at the expense of a higher-variance gradient, which the critic baseline and the conservative PPO clipping serve to attenuate. The entropy coefficient β is annealed linearly from β_0 to a floor $\beta_{\min}$ over the early training phase, so exploration dominates at first and gradually gives way to exploitation, and the learning rates follow their own decay.

Joint training is preceded by a supervised warm-up of the assignment and phase actors. The assignment actor is fitted by cross-entropy to a reference routing and the phase actor to the closed-form alignment evaluated for that same routing, the two being warmed jointly rather than separately so that the phase actor is not fitted to a routing the assignment actor will immediately abandon. The two stages are budgeted independently, a circuit evaluation over the warm-up buffer being substantially more expensive than the corresponding classical fit.

The assignment, phase, and common-rate actors each carry an *oracle teacher*, the term $\mathcal{L}_{\text{sup}}^j$ of (25), in which a_j^* is a per-state reference action and each factor is conditioned as in (21). Each reference is available at negligible cost, the phase target being the closed-form per-element alignment of Section IV-D, the assignment target a greedy routing solution, and the common-rate target the closed-form demand-filling share within each group. The teacher supplements the policy gradient rather than supplanting it, so a viable action remains available while the assignment distribution is still diffuse over the combinatorial space.

The power actor additionally applies a *fairness blend*. Rather than mixing the executed allocation, which would separate the executed action from the scored one, the blend acts on the Dirichlet concentration, $\boldsymbol{\alpha}_{\text{eff}} = (1 - a) \boldsymbol{\alpha} + a \left(\sum_i \alpha_i \right) \mathbf{t}$, so the mean allocation becomes a convex combination of the learned one and a target $\mathbf{t}$. The target is a fixed private share for the common-private split and the uniform vector for

the two within-share allocations. The blend weight a therefore sets how far the policy may concentrate power.

The classical encoder and policy layer of the quantum actor, all three sub-actors, and the critic are updated by standard back-propagation, and every module uses the Adam optimizer with per-module learning rates and gradient-norm clipping.

TABLE II: Summary of Main Notations

Symbol	Description	Symbol	Description
$\mathcal{K}$	Set of ground users	$\mathcal{M}$	Set of IRSs
K	Number of ground users	M	Number of IRSs
$\mathcal{N}$	Set of IRS reflecting elements	N	Number of reflecting elements
φ_k	IRS-selection variable of user k	$\mathcal{G}$	Set of active transmission groups
$\mathcal{K}_g$	Set of users assigned to group g	P_S	Maximum transmit power of the satellite
s_k	Original message intended for user k	$s_{c,g}$	Common stream of group g
$s_{k,p,g}$	Private stream of user k in group g	Φ_m	Phase-shift matrix of IRS m
$w_{c,g}$	Scalar precoding coefficient for $s_{c,g}$	$w_{k,p,g}$	Scalar precoding coefficient for $s_{k,p,g}$
$\mathbf{w}$	Set of scalar precoding coefficients	h	Effective channel coefficient
$\theta_{m,n}$	Phase shift of the n -th IRS element	$\mathcal{Q}_\theta$	Quantized phase-angle set
B	Number of phase quantization bits	g^{SU}, g^{SR}, g^{RU}	True channel coefficients of each link
$\hat{g}$	Estimated channel coefficient	Δg	CSI estimation error
x	Transmitted signal at the satellite	y_k	Received signal at user k
n_0	Additive white Gaussian noise	σ^2	Noise variance
γ	SINR for decoding the corresponding stream	$R_{k,c,g}$	Common-stream decoding rate of user k
$R_{c,g}$	Decodable common rate of group g	$C_{k,g}$	Allocated common rate for user k
$R_{k,p,g}$	Achievable private rate of user k in group g	$\epsilon^{SU}, \epsilon^{SR}, \epsilon^{RU}$	Rain-attenuation coefficient of each link
R_k	Total achievable rate of user k	R_{tot}	Achievable system sum-rate
$R_k^{\min}$	Minimum QoS requirement of user k	ϵ_o	Outage level of the QoS chance constraint
γ_d	RL discount factor	λ_{GAE}	GAE bias-variance parameter
ϵ_c	PPO clipping range	ϵ_q	QoS-penalty guard constant
λ_D	QoS-penalty weight	β	Entropy regularization coefficient
α_{AE}	Autoencoder loss weight	G_t	GAE λ -return regression target
ρ_t	PPO importance-sampling ratio	$\hat{A}_t$	Generalized advantage estimate
$V(\cdot)$	State-value function (critic)	η	VQC trainable parameters
c_s	SPSA perturbation step	τ	SoftmaxPQC inverse temperature (policy layer)
$\mathbf{s}_t$	State: affinity feature vector	$\mathbf{a}_t$	Action ($\varphi, \Phi, \mathbf{w}, \mathbf{C}$)
$\mathbf{z}_t$	AE latent vector ($\in \mathbb{R}^{2n_q}$)	$\bar{\mathbf{s}}_t$	Group-normalized affinity state
$a_{k,m}$	IRS path quality of user k via IRS m	q_m	Mean affinity of IRS m over users
p_m	Received satellite-to-IRS power	$\hat{\mathbf{s}}_t$	AE reconstruction of $\mathbf{s}_t$
$\mathcal{U}, \mathcal{R}$	User-qubit and IRS-qubit index sets	$\mathbf{W}_{\text{proj}}$	AE user-branch projection matrix
$\mathcal{L}_{AE}$	AE reconstruction loss	L	Number of VQC variational layers
n_q	Number of qubits	N_Q	Number of readout observables
$\hat{\mathbf{o}}_t$	VQC measurement (readout) vector	$\gamma, \mathbf{b}_{\text{LN}}$	Encoding LayerNorm learnable affine
λ_y, λ_z	Trainable encoding scales	ε	CSI error variable, $\varepsilon \sim \mathcal{CN}(0, 1)$
κ	CSI error coefficient	R_{LoS}	Satellite line-of-sight ground radius
ς_k	Rate shortfall of user k	$\mathbf{w}_p, \mathbf{w}_c$	Private and common power vectors
$\boldsymbol{\theta}$	VQC variational angles (component of η)	S_{ev}	Measurement shots at inference
S_{tr}	Measurement shots during training	$\gamma, \mathbf{b}_{\text{LN}}$	Encoding LayerNorm learnable affine
ϵ_{LN}	LayerNorm numerical-stability constant	Ω_{sf}	Average shadow-fading power gain
β_{blk}	Blocking factor on the direct link		

V. EXPERIMENT

A. Simulation Setup and Evaluation Protocol

The framework is evaluated at two network scales, $(K, M) = (5, 1)$ and $(10, 2)$, under the channel model of Section III. Table IV lists the simulation and training parameters. The two scales share every value in it except the user and panel counts and the circuit depth, $L = 2$ and $L = 3$ respectively.

Four baselines are used. AO re-solves every state by alternating optimization over the same four variables and is reported at the cheapest setting reaching its objective plateau. Greedy routes each user to the panel with the strongest estimated path and divides power evenly; Random draws every variable uniformly. PPO-flat ports the flat policy of [12], one network emitting all four variables from the unaggregated channel under a single PPO ratio. Only that architectural choice is taken from the original. It is given GAE advantages where [12] uses a plain Monte-Carlo return, an annealed entropy bonus it does not have, KL early stopping without which most samples clip and the gradient vanishes, the same critic and PopArt normalisation as the hierarchical actors, and a state augmented with the per-user demand and blockage indicators their system has no counterpart for. Its learning rate was re-tuned from their 10^{-4} to $3 \cdot 10^{-4}$ and swept again at $(10, 2)$, where it fails at both, and at 305K and 725K parameters it is the widest network of any arm.

Every learned agent performance is a mean over five training seeds. The checkpoint of a run is the one maximizing episode return subject to a 90% QoS floor, selected on three validation environments and re-scored on three disjoint test environments with actions taken greedily. The floor keeps the methods comparable, since PPO-flat trades service for rate monotonically through training whereas the hierarchical actors do not, so an unconstrained argmax would place each method at a different kind of operating point. At the extreme power budgets of Table VIII no checkpoint of any arm clears the floor, and the selection falls back there to unconstrained maximum return.

B. Sum-Rate and QoS Performance

Fig. 5 shows the training trajectories and Table V the corresponding greedy-inference results. Rows are compared on reward rather than on R_{tot} , since the policies settle at different QoS levels and a sum-rate obtained by underserving users is not comparable with one that meets the demand.

TABLE III: Hardware and software platform.

Component	Specification
CPU	Intel Core i7-9700K @ 3.60 GHz (8 threads)
GPU	NVIDIA GeForce RTX 4090 (24 GB)
RAM	64 GB
OS	Ubuntu 22.04.5 LTS
Python	3.11.15
Framework	NumPy 2.4.5, SciPy 1.17.1
GPU backend	CuPy 14.0.1 (CUDA runtime 12.9)

TABLE IV: Simulation and Training Hyperparameters

Parameter	Value	Parameter	Value
<i>System, channel, and quantum actor</i>		<i>Reinforcement learning and training</i>	
Users K / IRS M (Case 1/2)	(5, 1) / (10, 2)	Discount factor γ_d	0.95
Elements per IRS N	24	GAE parameter λ_{GAE}	0.9
Satellite power P_S	50 dBm	PPO clip ϵ_c / epochs	0.2 / 6
Carrier frequency f / altitude h_{SR}	1.58 GHz / 800 km	Rollout episodes / mini-batch	12 / 64
Antenna gains G_S, G_U	60 dBi	Entropy $\beta_0 \rightarrow \beta_{\min}$	$10^{-3} \rightarrow 3 \times 10^{-4}$
Noise n_0 (mean, variance)	(−30, 10) dBW	QoS penalty λ_D / guard ϵ_q	$1.5 / 10^{-3}$
Rain attenuation (μ, σ^2)	(4 dB, 0.1)	Reward noise averaging	16
CSI error coefficient κ	0.05	LR (AE ω / VQC η / layer ξ)	$10^{-4} / 10^{-4} / 3 \times 10^{-4}$
Phase quantization bits B	2 (4 levels)	LR (phase, power / C_k / critic)	$10^{-4} / 5 \times 10^{-5} / 3 \times 10^{-4}$
QoS demand $R_k^{\min}$	0.05 bps/Hz	Episodes / steps per episode	20,000 / 200
LoS radius R_{LoS} / IRS height	0.5 km / 20 m	Optimizer	Adam
Blocking factor β_{blk}	0.1	Gradient-norm clip (AE / actors)	0.1 / 0.5
User speed	1.5 m/s		
Qubits n_q / latent dim $2n_q$	8 / 16 (DNN latent 24)		
Variational layers L (Case 1/2)	2 / 3		
Data re-uploading	enabled		
SPSA step c_s / repetitions	0.1 / 16		
Measurement shots S (train / eval)	1500 / 3000		
Autoencoder loss weight α_{AE}	0.5		

Against the per-state solver, which attains 100% QoS by construction, both learned actors gain about 31% in reward at (5, 1) and 2.2% at (10, 2), and both clear the strongest heuristic by roughly 59% and 20%. The margin over AO therefore narrows sharply with scale. At (10, 2) the quantum actor’s 337.3 ± 8.8 lies within one standard deviation of the solver’s 330.1, so only the classical instantiation separates from AO there with confidence.

The two instantiations perform closely at both scales, their reward gap staying well inside the spread across training seeds. This is what the prior for parametrized quantum policies anticipates, namely that they match rather than exceed classical ones in task performance [17], [52]. Where they differ is in how

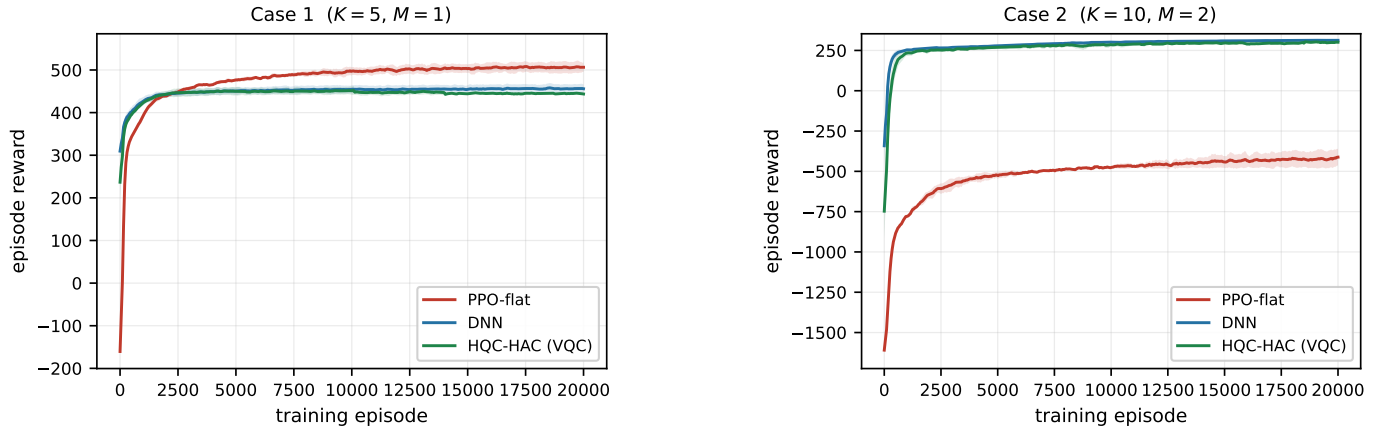


Fig. 5: Training curves for both problem sizes, showing mean episode return over five training seeds under stochastic actions, with shaded regions indicating one standard deviation. Resumed training runs are stitched across segments; for $(K, M) = (5, 1)$, the VQC seed budget decreases after 10,000 episodes as completed seeds are excluded from the mean, while PPO-flat at $(10, 2)$ averages two runs. The curves are not directly comparable with Table V, which reports greedy inference at checkpoints selected under a 90% QoS floor.

service is distributed. At $(10, 2)$ the quantum actor holds $94.4 \pm 1.4\%$ QoS against $92.1 \pm 2.0\%$, the only separation in this study exceeding two standard errors.

PPO-flat is the uneven one, competitive at $(5, 1)$ where it reaches 481.2 ± 6.4 at 96.0% QoS, the highest of any learned policy, yet falling at $(10, 2)$ to 0.557 bps/Hz and 47.2%. Section V-G identifies the mechanism behind that collapse.

C. Deployment Cost

Table VI decomposes both actors by role. The comparison is meaningful at the level of the whole assignment actor, since the circuit reaches its logits through a softmax layer that the classical policy network does not require, so the 80 and 96 parameters of the quantum core are not by themselves a substitute for the 1,410 and 2,230 of that network. Counted over the whole actor at inference, where the encoder is included and the reconstruction decoder discarded, the quantum pipeline is $9.2\times$ and $6.0\times$ smaller, and the complete policy is $18.4\times$ and $33.2\times$ smaller than the flat baseline.

Table VII reports what a decision and a training episode cost. At inference the solver spends 23.93 and 53.74 ms per decision against between 0.42 and 2.78 ms for the learned policies, but the more durable statement is the invocation count. AO calls the rate kernel 482 and 896 times per action, growing with the user population, whereas every learned policy calls it three times irrespective of scale. That count

TABLE V: Performance under greedy inference at $P_S = 50$ dBm. All policies are evaluated on the same 600 test states (3 environment seeds $\times$ 200 steps). Checkpoints are selected on disjoint validation environments by maximizing greedy episode reward subject to a 90% QoS requirement. Learned rows are mean $\pm$ standard deviation over five independent training seeds, the solvers being deterministic.

Policy	Reward	R_{tot}	QoS	Shortfall
<i>Case 1</i> ($K=5, M=1$)				
HQC-HAC (VQC)	490.3 ± 22.1	2.482 ± 0.100	$92.4 \pm 3.2\%$	$1.29\%/6.5 \cdot 10^{-4}$
DNN	494.8 ± 24.3	2.501 ± 0.121	$94.6 \pm 2.7\%$	$1.00\%/5.0 \cdot 10^{-4}$
PPO-flat	481.2 ± 6.4	2.417 ± 0.035	$96.0 \pm 3.8\%$	$0.23\%/1.1 \cdot 10^{-4}$
AO	375.1	1.876	100.0%	0.00%/0
Greedy	309.1	1.549	99.8%	$0.06\%/0.3 \cdot 10^{-4}$
Random	-307.6	0.579	47.8%	$33.81\%/169 \cdot 10^{-4}$
<i>Case 2</i> ($K=10, M=2$)				
HQC-HAC (VQC)	337.3 ± 8.8	1.742 ± 0.021	$94.4 \pm 1.4\%$	$0.88\%/4.4 \cdot 10^{-4}$
DNN	337.5 ± 2.3	1.747 ± 0.011	$92.1 \pm 2.0\%$	$1.08\%/5.4 \cdot 10^{-4}$
PPO-flat [†]	-367.1 ± 53.7	0.557 ± 0.039	$47.2 \pm 3.1\%$	$23.42\%/117 \cdot 10^{-4}$
AO	330.1	1.651	100.0%	0.00%/0
Greedy	281.2	1.434	99.4%	$0.21\%/1.1 \cdot 10^{-4}$
Random	-1582.9	0.315	19.6%	$63.90\%/320 \cdot 10^{-4}$

[†]Two runs at learning rates $3 \cdot 10^{-4}$ and $1 \cdot 10^{-4}$ rather than a five-seed mean.

is independent of language, machine, and, for the quantum actor, of the fact that its circuit is simulated rather than executed.

The quantum actor requires 18.4 and 25.2 s per episode against 1.7 and 2.7 for its classical counterpart and under 1.2 for the flat baseline, and issues about $3.9 \cdot 10^4$ circuit evaluations per episode. The solver requires no training whatsoever, so its entire cost falls at inference. That asymmetry is precisely the trade an amortized policy makes, and it is favourable only where the deployed system decides far more often than it is retrained.

Two measurements delimit where the parameter advantage originates. Reducing the register from eight qubits to six leaves performance inside the spread across training seeds, and measuring only single-qubit Z observables, eight of them in place of the 26 of the structured readout, does the same. The advantage therefore rests on how the state is compressed for the circuit rather than on the circuit's expressive degrees of freedom, which Section V-E examines in detail.

TABLE VI: Parameter budget across problem sizes (K, M) . The upper block reports the *assignment actor*, where the VQC and DNN variants are directly substitutable; training parameters additionally include the autoencoder decoder, which is discarded after training. The lower block reports the complete inference policy, the level at which PPO-flat is comparable because it jointly outputs all four decision variables.

	(5, 1)	(10, 2)
<i>Assignment actor</i>		
VQC core	80	96
Softmax policy layer	391	1,291
AE encoder	272	446
AE decoder (<i>training only</i>)	561	2,268
VQC, inference	743	1,833
VQC, training	1,304	4,101
DNN	6,810	11,086
DNN / VQC (inference / training)	$9.2 \times / 5.2 \times$	$6.0 \times / 2.7 \times$
<i>Complete policy, inference</i>		
HQC-HAC (VQC)	16,538	21,831
Classical (DNN)	23,629	32,108
PPO-flat	305,016	725,239
PPO-flat / HQC-HAC	$18.4 \times$	$33.2 \times$

TABLE VII: Cost of one decision at inference and one training episode. The upper half reports per-decision inference cost, while the lower half reports per-episode training cost. Within each half, the first block gives wall-clock time and the second gives implementation-independent invocation counts. Wall-clock values are medians. AO is re-solved independently for each state and is not trained; hence, training entries do not apply.

		AO	PPO-flat	DNN	VQC
<i>Inference, per decision</i>					
Latency (ms)	(5, 1)	23.93	0.42	0.60	1.94
	(10, 2)	53.74	0.66	0.85	2.78
Rate-kernel evals	(5, 1)	482	3	3	3
	(10, 2)	896	3	3	3
<i>Training, per episode</i>					
Wall-clock (s)	(5, 1)	n/a	0.87	1.72	18.38
	(10, 2)	n/a	1.19	2.72	25.23
Circuit evals	(5, 1)	n/a	n/a	n/a	$3.9 \cdot 10^4$
	(10, 2)	n/a	n/a	n/a	$3.9 \cdot 10^4$

D. Robustness to Distribution Shift

Table VIII evaluates every policy under thirteen shifts applied after training. Quadrupling the CSI error from $\kappa = 0.05$ to 0.20 costs the quantum actor 0.04% of its sum-rate at (5, 1) and 0.7% at (10, 2), and raising the noise variance from 10 to 16 dB² costs 1.1% and 2.3%. Both learned actors therefore degrade gracefully under exactly the perturbations that imperfect CSI is meant to model.

TABLE VIII: Zero-shot generalization under distribution shifts for both cases. Policies are trained at the base operating point ($P_S = 50$ dBm, $\kappa = 0.05$, $\sigma^2 = 10$ dBW, user speed 1.5 m/s) and evaluated without retraining under shifted environments, while AO is re-optimized for each shifted state. Combined-A denotes $\{P_S = 40$ dBm, $\kappa = 0.10$, speed = 3 m/s} and Combined-B denotes $\{P_S = 60$ dBm, $\kappa = 0.20$, speed = 6 m/s, $\sigma^2 = 16$ dBW}. Operating points with $P_S \neq 50$ dBm are retrained and reported separately, as transmit power is a deployment parameter rather than a runtime environmental variable. The N -shift rows are zero-shot for the hierarchical actors, whose affinity map and phase sub-actor do not depend explicitly on N ; PPO-flat is retrained because its unaggregated channel input dimension changes with N . The 90% QoS floor is applied only to base-row checkpoint selection; shifted rows use unconstrained argmax return. A dagger indicates QoS below 90%, for which the corresponding rate is not directly comparable with entries satisfying the QoS floor.

Shift	AO			PPO-flat			DNN			HQC-HAC (VQC)		
	R_{tot}	QoS	Reward	R_{tot}	QoS	Reward	R_{tot}	QoS	Reward	R_{tot}	QoS	Reward
<i>Case 1 ($K = 5, M = 1$)</i>												
Base	1.8366	100.0%	367.3	2.4658	93.6%	489.3	2.4829	94.4%	491.1	2.4653	92.3%	486.5
$P_S = 40$	1.3278	100.0%	265.3	1.0812	92.7%	197.0	1.8350	76.9% [†]	329.1	1.7917	78.8% [†]	321.5
$P_S = 60$	1.9928	100.0%	398.6	2.9638	97.5%	587.3	2.8024	94.3%	556.5	2.7942	95.3%	556.3
$P_S = 70$	2.0809	100.0%	416.2	10.2956	99.9%	2059.0	3.8151	81.7% [†]	596.2	3.9302	80.5% [†]	601.8
$N = 16$	1.8216	100.0%	364.3	2.1698	92.5%	430.0	2.3710	93.2%	466.3	2.3449	92.1%	461.9
$N = 32$	1.9105	100.0%	382.1	2.6684	90.4%	531.5	2.5764	96.0%	512.2	2.5720	92.9%	508.8
$\kappa = 0.10$	1.8369	100.0%	367.4	2.4667	93.6%	489.5	2.4808	94.3%	490.2	2.4661	92.3%	486.7
$\kappa = 0.15$	1.8365	100.0%	367.3	2.4662	93.5%	489.4	2.4722	94.1%	487.1	2.4665	92.2%	486.5
$\kappa = 0.20$	1.8366	100.0%	367.3	2.4659	93.5%	489.2	2.4588	93.0%	478.7	2.4654	92.2%	486.3
$\sigma^2 = 13$	1.8232	100.0%	364.6	2.4425	92.9%	484.1	2.4664	94.0%	487.4	2.4507	92.0%	483.0
$\sigma^2 = 16$	1.8104	100.0%	362.1	2.4203	92.1%	478.8	2.4506	93.7%	483.6	2.4337	91.7%	479.0
Combined-A	1.1427	98.9%	225.2	1.0048	91.2%	173.2	1.7265	78.0% [†]	308.8	1.7431	76.7% [†]	307.0
Combined-B	1.7818	100.0%	356.3	2.5592	97.4%	504.7	2.8937	95.4%	574.6	2.8356	95.4%	564.5
<i>Case 2 ($K = 10, M = 2$)</i>												
Base	1.6515	100.0%	330.3	0.5874	54.4% [†]	−326.5	1.7522	91.5%	337.4	1.7397	94.4%	337.1
$P_S = 40$	1.2979	93.8%	240.1	0.8579	76.3% [†]	−244.1	0.9825	92.7%	162.0	0.9786	93.2%	162.3
$P_S = 60$	1.7418	100.0%	348.4	2.6097	89.7% [†]	503.8	2.2359	87.4% [†]	438.7	2.1907	92.0%	432.9
$P_S = 70$	1.7581	100.0%	351.6	10.1880	98.0%	2036.7	2.2988	87.7% [†]	454.3	2.2860	91.1%	453.3
$N = 16$	1.6341	100.0%	326.8	0.5771	51.5% [†]	−283.0	1.6688	89.8% [†]	309.5	1.6594	92.6%	308.6
$N = 32$	1.6575	100.0%	331.5	0.5138	43.5% [†]	−397.4	1.7754	93.1%	345.4	1.7568	95.3%	345.1
$\kappa = 0.10$	1.6513	100.0%	330.3	0.5874	54.4% [†]	−326.4	1.7516	91.5%	337.4	1.7380	94.3%	336.3
$\kappa = 0.15$	1.6513	100.0%	330.3	0.5873	54.4% [†]	−326.5	1.7495	91.5%	336.2	1.7321	94.2%	334.5
$\kappa = 0.20$	1.6508	100.0%	330.2	0.5872	54.4% [†]	−326.4	1.7478	91.4%	335.5	1.7239	94.0%	331.9
$\sigma^2 = 13$	1.6410	100.0%	328.2	0.5909	55.3% [†]	−337.4	1.7321	91.2%	332.3	1.7157	94.2%	331.4
$\sigma^2 = 16$	1.6309	100.0%	326.2	0.5943	56.0% [†]	−347.2	1.7134	90.8%	327.3	1.6942	93.9%	326.3
Combined-A	1.0949	93.6%	201.5	0.8163	72.8% [†]	−317.3	0.9646	91.7%	159.2	0.9632	92.3%	158.4
Combined-B	1.6233	100.0%	324.4	2.7734	84.2% [†]	503.3	2.3042	89.8% [†]	453.0	2.2558	93.5%	445.1

Transmit power is a different axis, a deployment parameter rather than a runtime one, so those rows are retrained at their own budget. Both learned actors retain an advantage down to 50 dBm, while at 40 dBm and under combined-A the solver leads on both metrics at once.

PPO-flat fails discontinuously. At (10, 2) it falls below the solver at ten of thirteen shifts, and its column is static across the base point and all five CSI and noise shifts, so the failure is one break at the base operating point rather than a response to the shifts. The same table nevertheless places it ahead of every

other method once the budget is generous, at 503.8 and 2036.7 against 438.7 and 454.3 for the classical actor at 60 and 70 dBm. The hierarchical actors fall below the solver only where the solver itself leaves 100% QoS. Daggers [†] mark operating points below the 90% floor, twenty-two of the seventy-eight learned entries. At (10, 2) the quantum actor clears it at all thirteen shifts and PPO-flat at one.

E. What the Quantum Resources Contribute

Table IX varies the decision core along three axes while holding the encoder and the policy layer fixed, so the parameter column counts the substitutable module alone on both sides. Reducing the register from eight qubits to six moves R_{tot} from 1.742 ± 0.021 to 1.735, inside the spread across training seeds, at a core of 72 parameters instead of 96. Removing the entangling bridges while holding the parameter count and the readout unchanged yields 1.703, again within the seed spread, although that run used a shorter training budget than the reference chain and the comparison should be read accordingly.

Table XI varies the measured observable set at a fixed register. The sets are not nested and their sizes do not rank them, so each case is reported separately. At (10, 2), measuring eight single-qubit Z observables reaches 1.749 at 95.9% QoS against 1.742 ± 0.021 at $94.4 \pm 1.4\%$ for the structured readout with 26.

Taken together these measurements locate the contribution. The autoencoder is indispensable, since a direct encoding would demand 17 and 44 qubits at the two scales and place state-vector simulation with per-step gradients out of reach. The register size, the observable set and the two-qubit correlations are, within the resolution of these experiments, inert.

F. Representation Quality

Table X asks how much assignment-relevant information each latent carries, independently of the policy that consumes it. A lightweight classical probe is trained to recover the oracle assignment from frozen representations on a common held-out state set, so the comparison isolates the encoding from the downstream actor. The quantum and classical encoders reach comparable accuracy, and both are close to what the raw state and the full channel afford. The encoders therefore concentrate rather than merely preserve the information the assignment depends on, and the residual gap between a probe and the live policy is one of credit assignment during training rather than of representational capacity.

TABLE IX: Ablation of the *decision core* at Case 2. All rows share the same autoencoder encoder and softmax policy layer, so the parameter column counts the substitutable decision module alone on both sides. The two reference rows are the Case-2 main-table entries; each block below varies one axis away from the quantum reference and omits the reference setting itself. Rows carrying a spread are five-training-seed means, the rest are single seeds.

Decision core	Configuration	Params	R_{tot}	QoS
<i>Reference</i>				
VQC	$n_q=8$, $L=3$, entangled	96	1.742 ± 0.021	$94.4 \pm 1.4\%$
Classical MLP	(40)	2,230	1.747 ± 0.011	$92.1 \pm 2.0\%$
<i>Classical width</i>				
	(256, 128)	—	—	—
	(128, 128)	—	—	—
	(64, 64)	—	—	—
	(64, 32)	—	—	—
	(32, 16)	—	—	—
<i>Circuit depth</i> ($n_q=8$)				
	$L = 1$	64	—	—
	$L = 2$	80	—	—
	$L = 4$	112	—	—
	$L = 6$	144	—	—
<i>Register size</i> ($L=3$)				
	$n_q = 6$	72	1.735	94.6%
	$n_q = 10$	120	—	—
<i>Entanglement</i> ($n_q=8$, $L=3$)				
	no CZ bridges	96	1.703	94.6%

TABLE X: Representation quality (Case 2): supervised recovery of the oracle assignment from frozen representations on a common held-out state set ($N = 384$). Both actors encode the identical states, and the resulting latent representations are evaluated independently of the downstream policy. Exact accuracy measures full assignment recovery, while IRS/direct accuracy measures correct classification into IRS-assisted versus direct transmission.

Representation	Exact acc.	IRS/direct acc.
$\mathbf{z}_t$ (VQC-AE)	96.2%	97.1%
$\mathbf{z}_t$ (DNN)	95.8%	96.8%
Raw state $\mathbf{s}_t$	95.8%	97.0%
Full channel	95.9%	97.1%

G. Budget Allocation Policy

The comparisons so far report what each policy achieves; Fig. 6 reports how, as the Lorenz curve of the private power across users at four transmit budgets. The solver lies on the diagonal at every rung, Gini 0.00 throughout, because it divides private power equally by construction. Whatever the learned policies

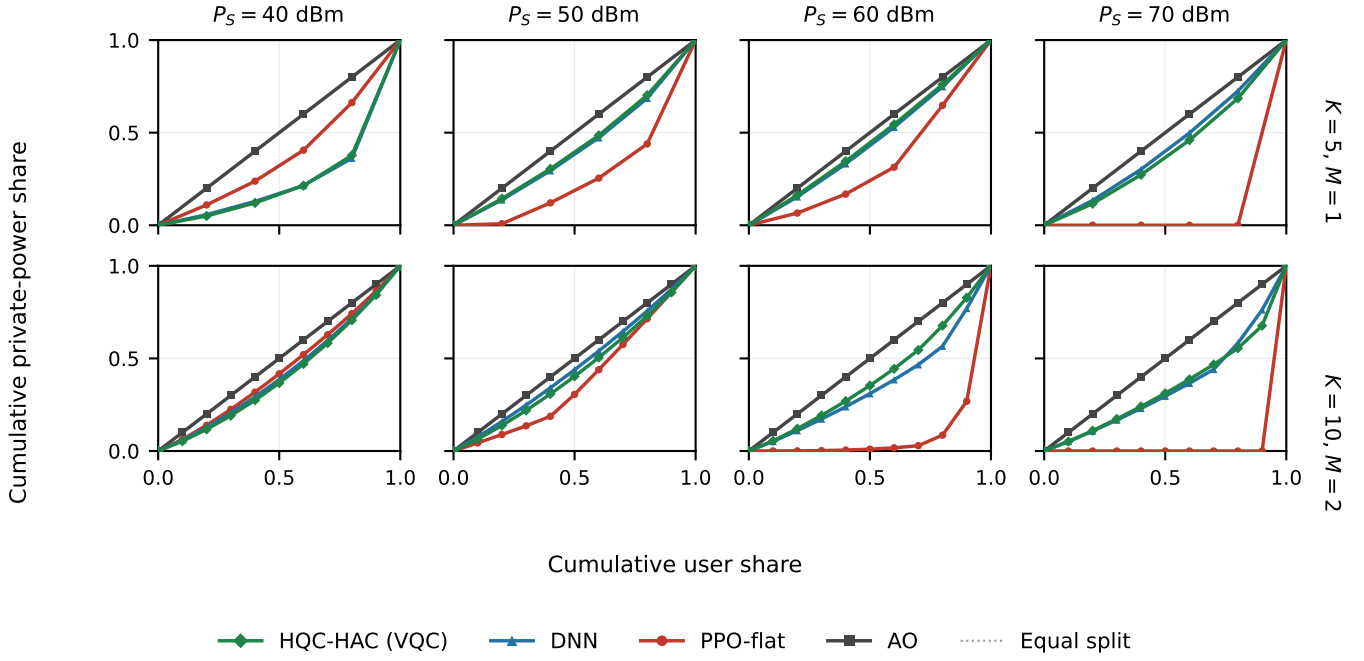


Fig. 6: Lorenz curves of private-power allocation across transmit-power budgets. Users are ordered by received private power; the diagonal denotes equal allocation, and the area between a curve and the diagonal gives the Gini coefficient. The common stream is omitted because it is allocated per group rather than per user. Curves use the greedy checkpoints of Table VI.

gain on the power axis of Section VI is gained here.

For the hierarchical actors the dependence on the budget is not monotone. Concentration attains a minimum at intermediate budgets, 0.10 and 0.07 at $(5, 1)$ and 60 dBm, and rises at both extremes, to 0.50 for either actor at 40 dBm and to 0.14 and 0.19 at 70 dBm. At the low end no even split brings any user to its demand, so serving a subset well is what the reward prefers; at the high end the shortfall penalty is capped at λ_D per withheld user while the rate term grows with P_S , so withholding eventually pays for itself. This is the trade-off of Section VI seen along the power budget rather than along λ_D .

PPO-flat shows only the second rise, and without bound. At 40 dBm it is the least concentrated of the three learned policies at both scales; at 70 dBm it places the entire private budget on one user. It answers the incentive that rewards concentration where the budget is large enough that the demand is met whatever the allocation, and not the one that punishes it where the budget is tight. That single mechanism accounts for both of its anomalies, the collapse at $(10, 2)$ in Table V and the lead at 60 and 70 dBm in Table VIII, each being the same policy answered by a different budget.

VI. RATE-QoS FRONTIER

The policies of Section V-B settle at different service levels, so the reference they are measured against has to be defined at a service level too. For a fixed environment let

$$F(q) = \max_{\varphi, \Phi, \mathbf{w}, \mathbf{C}} \mathbb{E}[R_{\text{tot}}] \quad \text{s.t.} \quad \mathbb{E}[\text{QoS}] \geq q, \quad (29)$$

the largest mean sum-rate attainable at mean QoS q under the constraints of (P0). The constraint binds the mean, as Table V reports it. Constraining every state separately defines a different and strictly lower quantity, since it forbids trading service between states, and evaluating one against the other is a common way to overstate how close a policy sits. We obtain F on the 600 test states of Table V by enumerating every assignment, solving the phase exactly for each, sweeping the power split on a grid, filling C_k by demand, and tracing the upper envelope by Lagrangian sweep, with decisions taken on $\hat{g}$ and scored on g as in training. Because the power search is a grid rather than a full optimization, F is an achievable lower bound on the true frontier.

In the high-power regime the SINRs reduce to $x_k/(1-x_k-y_{g(k)})$ and $y_g/(1-y_g)$ in the power fractions $x_k = w_{p,k}/P_S$ and $y_g = w_c^g/P_S$: the channel magnitudes cancel and the frontier is set by how many users are served and how the unit budget is divided. Under an equal split across m served users the private sum-rate is the concentration identity $R(m) = m \log_2 \frac{m}{m-1}$, decreasing in m with floor $1/\ln 2$.

Fig. 7 shows the frontier is shallow across the high-service region and steepens only once QoS is allowed to fall, the signature of an interference-limited link, in which concentrating power raises one user's rate and everyone else's interference with it. Serving every user is not merely costly but sometimes infeasible, attainable in 99.5% of states at (5, 1) and in 87.3% at (10, 2). Of the 3.38 and 2.15 bps/Hz the frontier reaches at full service, the equal-split identity accounts for 1.61 and 1.52; the remainder is bought by dividing power unequally, the axis on which a solver that splits private power evenly by construction cannot compete.

All four policies lie well below the frontier, and their distances from it are not read the same way. The solver is pinned to the right-hand edge, because its common-rate allocation is ordered by service and it therefore ends at full QoS whatever the channel, and that edge is where the frontier is lowest. Proximity to F is accordingly not the figure of merit, since F maximizes rate at a fixed service level whereas every

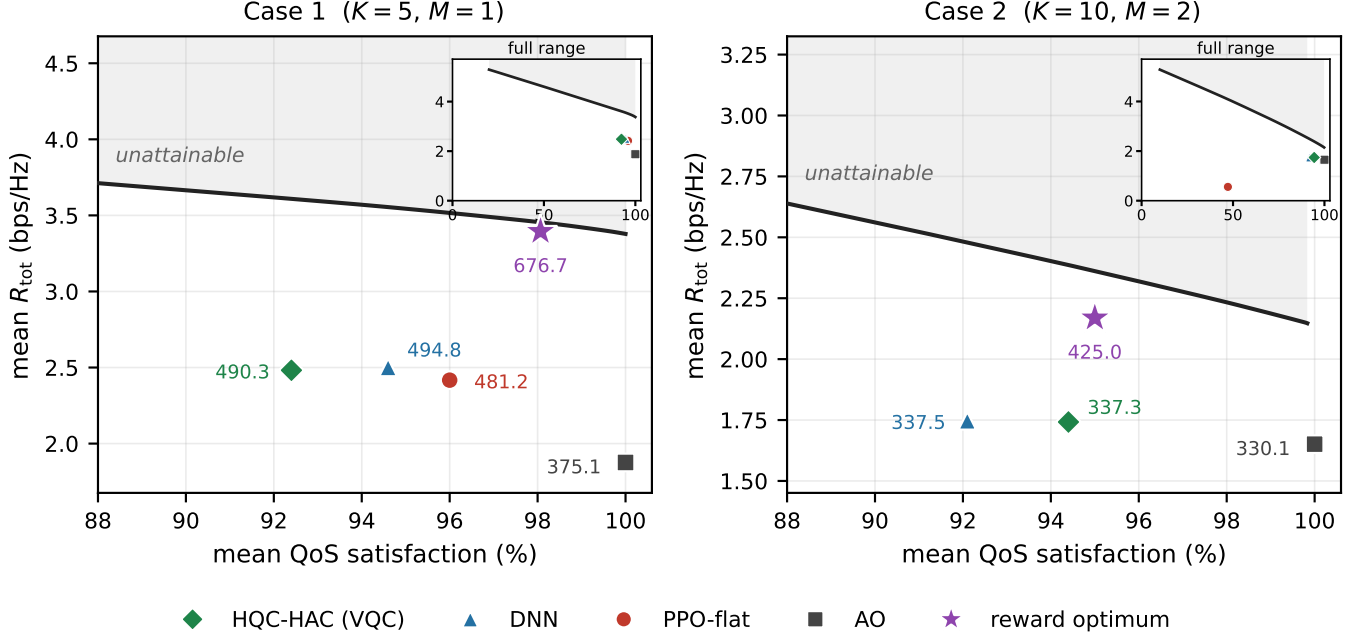


Fig. 7: Rate–QoS frontier and policy operating points for 600 test states per case. The frontier in (22) is obtained by a Lagrangian sweep over the achievable sets; since assignment is exhaustively enumerated and phase is solved exactly, while power splitting is grid-swept, it provides an achievable lower bound, with the shaded region unattainable under this construction. Markers denote the greedy-inference operating points from Table III, annotated by episode return, while stars indicate the reward-optimal operating points over the same achievable sets. The inset shows the full QoS range for Case 2, including PPO-flat.

policy here maximizes the QoS-penalized reward of (22). The two targets are distinct, and the point the reward itself selects lies at 98.1% and 95.0% QoS rather than at 100%. Measured on that objective the learned actors reach 73.1% and 79.4% of the attainable return, against 55.4% and 77.7% for the solver.

The operating point is therefore chosen by λ_D rather than arrived at by accident. Withholding service from one user costs $\lambda_D \hat{p}$ with $\hat{p} = (R_k^{\min} / (R_k^{\min} + \epsilon_q))^2$, so the serve-all optimum takes over above

$$\lambda_{\text{crit}} = \frac{V_{\text{rate}}^* - R^*(K)}{(K-1)\hat{p}}, \quad V_{\text{rate}}^* = \mathbb{E}[\log_2(1 + \Gamma_{\max})], \quad (30)$$

V_{rate}^* being the rate of placing the whole budget on the strongest link, the one configuration free of intra-system interference. Measured, $\lambda_{\text{crit}} = 1.74$ at $(5, 1)$ and 1.01 at $(10, 2)$, which places the training value $\lambda_D = 1.5$ on opposite sides of the two cases. At the smaller scale the reward’s global optimum is the degenerate single-user concentration, and entropy regularization is what holds the policy in the serve-all region; at the larger scale the reward already prefers serving everyone.

That the reward optimum sits below full service follows from the penalty rather than from any shortage

of power. Its marginal cost $2\lambda_D t / (R_k^{\min} + \epsilon_q)^2$ vanishes as the shortfall $t \rightarrow 0$, so the reward-optimal common-rate split is a water-filling solution, the shares solving $\sum_{k \in \mathcal{K}_g} (\zeta_k - t^*)^+ = R_{c,g}$ for a common level t^* , which settles the weakest users marginally below their demand. Full service and reward optimality are therefore separate targets at every λ_D , and the comparison in Fig. 7 is made at matched service for that reason.

VII. DISCUSSION

The evidence assembled here supports a claim of feasibility and efficiency rather than of quantum advantage on the task itself. At both scales the variational actor and the classical actor it replaces are separated by less than a quarter of the spread across training seeds, by 4.5 against a spread near 22 at (5, 1) and by 0.2 against a spread near 9 at (10, 2), which is what the prevailing prior for parametrized quantum policies anticipates [17], [52]. What distinguishes them is the budget at which that performance is obtained. The substitutable decision core is 80 and 96 parameters against the 1,410 and 2,230 of the network it replaces, and the complete policy is $18.4\times$ and $33.2\times$ smaller than the flat baseline. Set against a solver re-run at every channel realization [8], the learned policy converts a recurring per-state cost into a single offline one and thereafter answers each realization with three evaluations of the rate kernel in place of several hundred.

The measurements also separate what the design owes to compression from what it owes to the circuit. The autoencoder is indispensable because it decouples the register from the problem size, a direct angle encoding of the affinity state requiring 17 and 44 qubits at the two scales and 81 at $K=15$ while the register stays at eight throughout. The circuit's own expressive degrees of freedom are, within the resolution of these experiments, inert. Reducing the register from eight qubits to six, and measuring eight single-qubit Z observables in place of the 26 of the structured readout, each leave the sum-rate inside the seed spread, as does removing the entangling bridges at a fixed parameter count (Section V-E). We report this as a negative result about the circuit rather than about the framework. It locates the parameter saving in how the state is compressed for the register rather than in the correlations the register is able to represent, and it suggests that the assignment problem at these scales is not one whose structure a shallow entangling ansatz exploits.

That saving carries a cost, and the regime in which it is repaid is narrow. Training the quantum actor costs 18.4 and 25.2 s per episode against 1.7 and 2.7 for its classical counterpart and issues roughly 3.9×10^4 circuit evaluations per episode, so the amortization argument holds only where a deployed system decides far more often than it is retrained. The margin over the solver is bounded in a second sense. It is widest where the power budget is generous and reverses at 40 dBm and under the combined shift, the two operating points at which the solver itself ceases to reach full service, so the learned policies fall behind where the problem becomes constrained rather than uniformly.

VIII. LIMITATIONS

The quantum actor is evolved by state-vector simulation, so gate noise, the sampling error remaining beyond the shot budget modelled here, and compilation onto a device topology all lie outside the measurement, and the inference figures are algorithmic rather than hardware results. An eight-qubit register is within reach of present devices, but the $\approx 3.9 \cdot 10^4$ circuit evaluations each training episode consumes would be device time rather than simulated time, and it is that budget, not the register width, that separates the framework from execution on physical hardware.

The parameter ratio depends on where the boundary between the quantum and classical parts is drawn, falling from $9.2 \times$ and $6.0 \times$ at the assignment actor, the level at which the two arms are interchangeable, to roughly $1.4 \times$ once the classical sub-actors they share are counted. Every learned row averages five training seeds, at which resolution the QoS separation at (10, 2) is the sole difference between the two instantiations exceeding two standard errors.

The system model is restricted in two further respects. A single-antenna satellite admits only scalar precoding, so the spatial degrees of freedom of a multibeam payload, and the inter-group interference management they would afford, lie outside the formulation. The policy is also trained for a fixed user population and panel count, and the study spans two network scales and one channel family, so the narrowing of the margin over the solver with scale is observed rather than established.

IX. CONCLUSION

We presented HQC-HAC, a hybrid quantum–classical hierarchical actor–critic for sum-rate maximization in IRS-assisted grouped-RSMA integrated satellite–terrestrial networks. The formulation casts the

joint assignment, phase, power, and common-rate problem as a sequential dependency chain under QoS, discrete-phase, blockage, and imperfect-CSI constraints. A variational quantum circuit resolves the combinatorial $(M+1)^K$ assignment, three classical sub-actors resolve the factorable low-level controls, and an autoencoder compresses the state so that the register size follows the design rather than the network size. Across two scales the method improves on the strongest heuristic by 59% and 20% in episode return and matches a classical actor of identical role, input, and output to within the spread across training seeds, using $9.2\times$ and $6.0\times$ fewer parameters in the actor the two share and serving each channel realization with three evaluations of the rate kernel rather than the several hundred a per-state solver requires. It exceeds that solver by 31% at the smaller scale, with the margin narrowing to within one standard deviation at the larger one. The ablations attribute the outcome to the compression rather than to the register size, the observable set, or the entangling structure, none of which alters performance measurably at these scales. Future work includes execution on quantum hardware, a multibeam payload, a user population that varies without retraining, and a systematic sweep of circuit depth against classical width at matched budgets.

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APPENDIX A

ARCHITECTURAL AND AUTOENCODER ABLATIONS

A. Autoencoder Latent Compression

The AE compresses the high-dimensional state into a latent $\mathbf{z}_t \in \mathbb{R}^{2n_q}$ that (a) is necessary to fit the input into the n_q -qubit register, and (b) concentrates, rather than merely preserves, the assignment-relevant information. We assess (i) the qubit cost of encoding without compression against the fixed $n_q=12$ with AE; and (ii) the information loss of the fixed-width latent as the problem scales, governed by the compression ratio $(K(M+2)+2M)/2n_q$, which grows in K and M (N is filtered out by the affinity construction).

B. VQC Architectural Choices

Five architectural choices define the quantum actor; each is varied one at a time against the reference configuration ($n_q=8$, $L=2$ and $L=3$ at the two scales, nearest-neighbor CZ chain with M cross-block bridges, $R_y R_z$ angle encoding with data re-uploading, R1 readout).

1) *Circuit Depth*: The number of variational layers L widens the accessible frequency spectrum of the re-uploading circuit [43], so performance is expected to improve with L up to saturation. The depth sweep is pending: —.

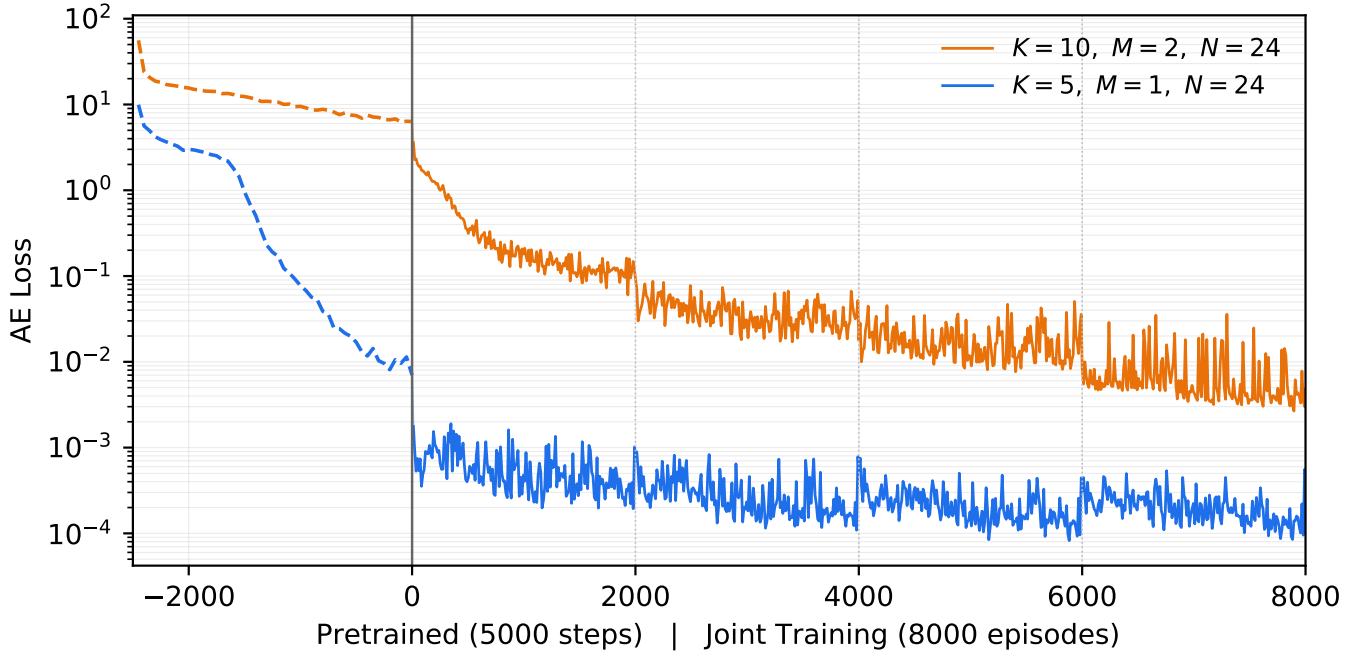


Fig. 8: AE reconstruction loss L_{AE} during autoencoder pre-training and subsequent joint RL training.

2) *Entanglement Pattern*: Candidate patterns of increasing connectivity — none, linear chain, ring, full — are compared against the reference chain augmented with M cross-block CZ bridges. The bridges are the problem-specific element: they couple the user- and IRS-qubit registers and are hypothesized to carry the user–IRS interaction that the assignment must model, whereas denser generic patterns add entanglement without this alignment. The toggle runs are pending: —.

3) *Qubit Count and Latent Dimension*: The register size n_q fixes the latent width $2n_q$ and hence the compression ratio $(K(M+2)+2M)/2n_q$; it trades representation capacity against the 2^{n_q} simulation cost. The register is held at $n_q=8$ throughout; reducing it to six leaves performance inside the spread across training seeds (Section V-E), and a wider sweep is pending: —.

4) *Feature Map*: The reference encoding loads each latent pair through $R_y R_z$ rotations with a tanh squash and re-applies the encoding block at every layer (data re-uploading [42]). Alternative feature maps (amplitude encoding, IQP-style maps) change the accessible function class and the loading cost. A comparison is pending: —.

5) *Readout Design*: The readout ladder varies the measured Pauli set at otherwise identical hyperparameters, from single-qubit Z alone to sets that add two-qubit correlations of differing reach. The sets are not nested and their sizes do not rank them, so Table XI reports each case separately rather than as

TABLE XI: Readout-design ablation, reported separately per case. All configurations are identical except the measured observable set, and every entry is a fresh run at $n_q=8$. $|\hat{\mathbf{o}}|$ is the number of observables the policy layer receives. The R1 rows are the five-training-seed main-table figures of Table V; the ablation rows are single seeds and carry no spread. The cross- ZZ set pairs every user qubit with every IRS qubit and adds nearest-neighbour pairs inside the user block; it is not the all-pairwise set, which would give 36 observables at $n_q=8$. R1 carries one more observable than cross- ZZ because it summarises each user’s coupling to all other users in a single averaged term rather than in nearest-neighbour pairs, so $|\hat{\mathbf{o}}|$ does not order the sets by how much they measure.

Readout	Observables ($ \hat{\mathbf{o}} $)	R_{tot}	QoS
<i>Case 1 ($K = 5, M = 1$)</i>			
single- Z	8	—	—
$Z + \text{NN-}ZZ$	15	—	—
$Z + \text{cross-}ZZ$	21	—	—
R1 (structured)	22	2.482 ± 0.100	$92.4 \pm 3.2\%$
<i>Case 2 ($K = 10, M = 2$)</i>			
single- Z	8	1.749	95.9%
$Z + \text{NN-}ZZ$	15	—	—
$Z + \text{cross-}ZZ$	25	—	—
R1 (structured)	26	1.742 ± 0.021	$94.4 \pm 1.4\%$

a monotone ladder.

C. Entropy Regularization and Annealing